



Speed and Distance Estimation

Continuous-Time Animal Movement Modeling

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Overview

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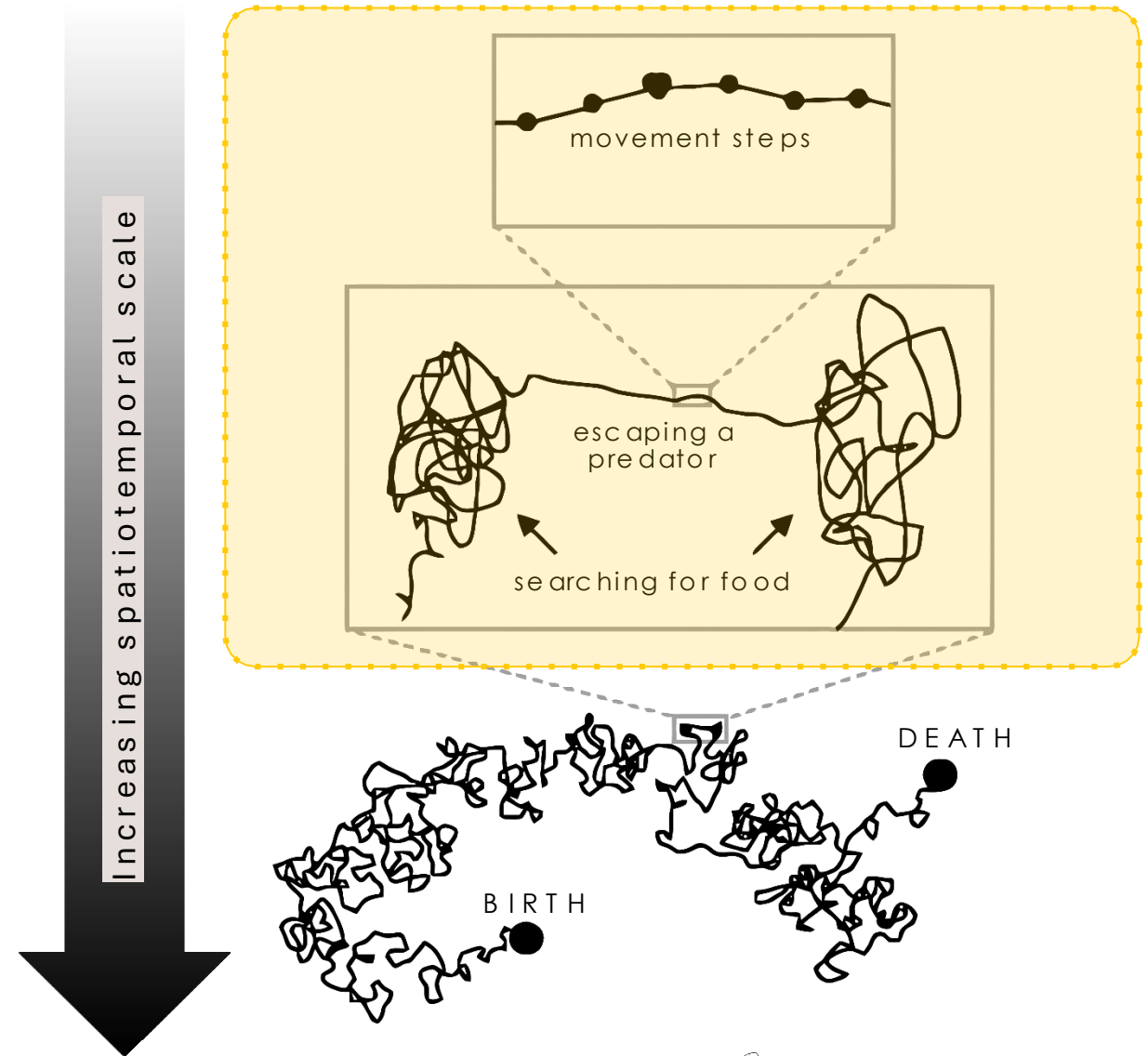
- 1) Estimating speed and distance
- 2) Continuous-time speed and distance
- 3) Applications
- 4) Limitations
- 5) Diffusion rate

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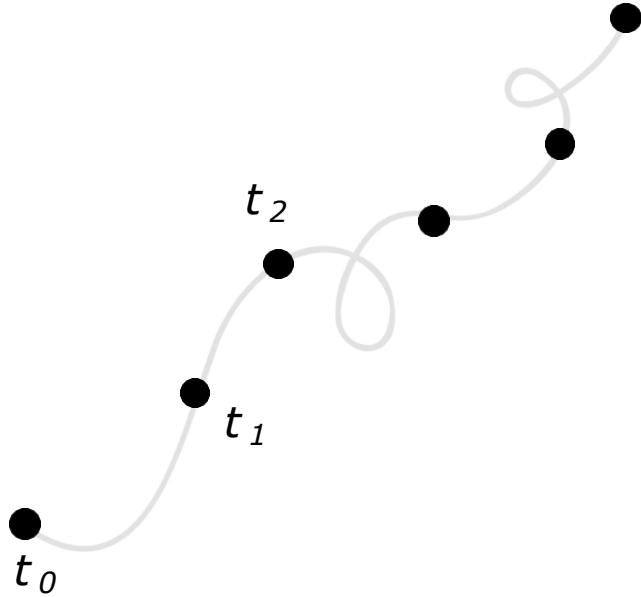
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Introduction

- Speed- and distance-related metrics provide quantifiable links between **behavior** and **energetics**,
 - Can inform on risk/reward tradeoffs or as signals of *anthropogenic disturbance*.



Estimating speed & distance

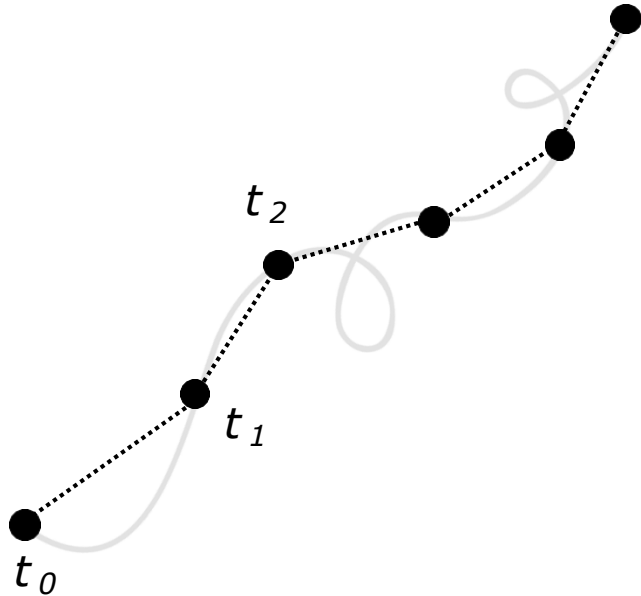


- *Speed and distance* are among the most routinely estimated metrics from animal tracking data.
- Usually estimated by summing the **straight-line distance (SLD)** between location estimates...

$$\hat{d} = |\Delta \mathbf{r}| = \sqrt{\Delta x^2 + \Delta y^2}$$

... and divide that by Δt if speed is the desired metric.

Estimating speed & distance

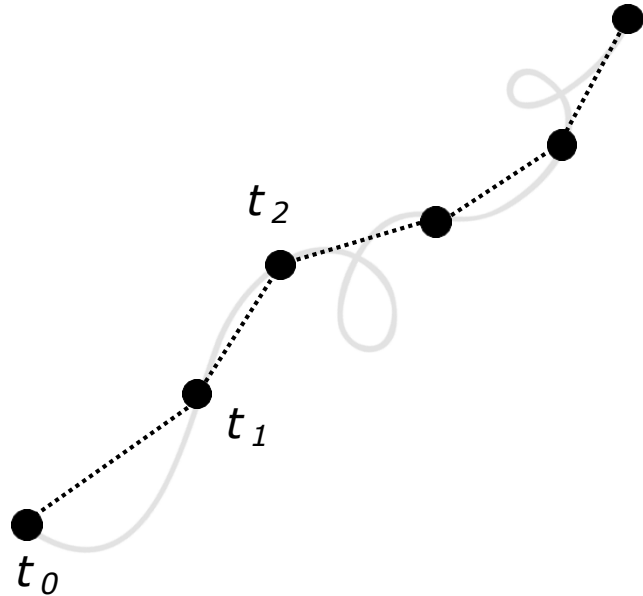


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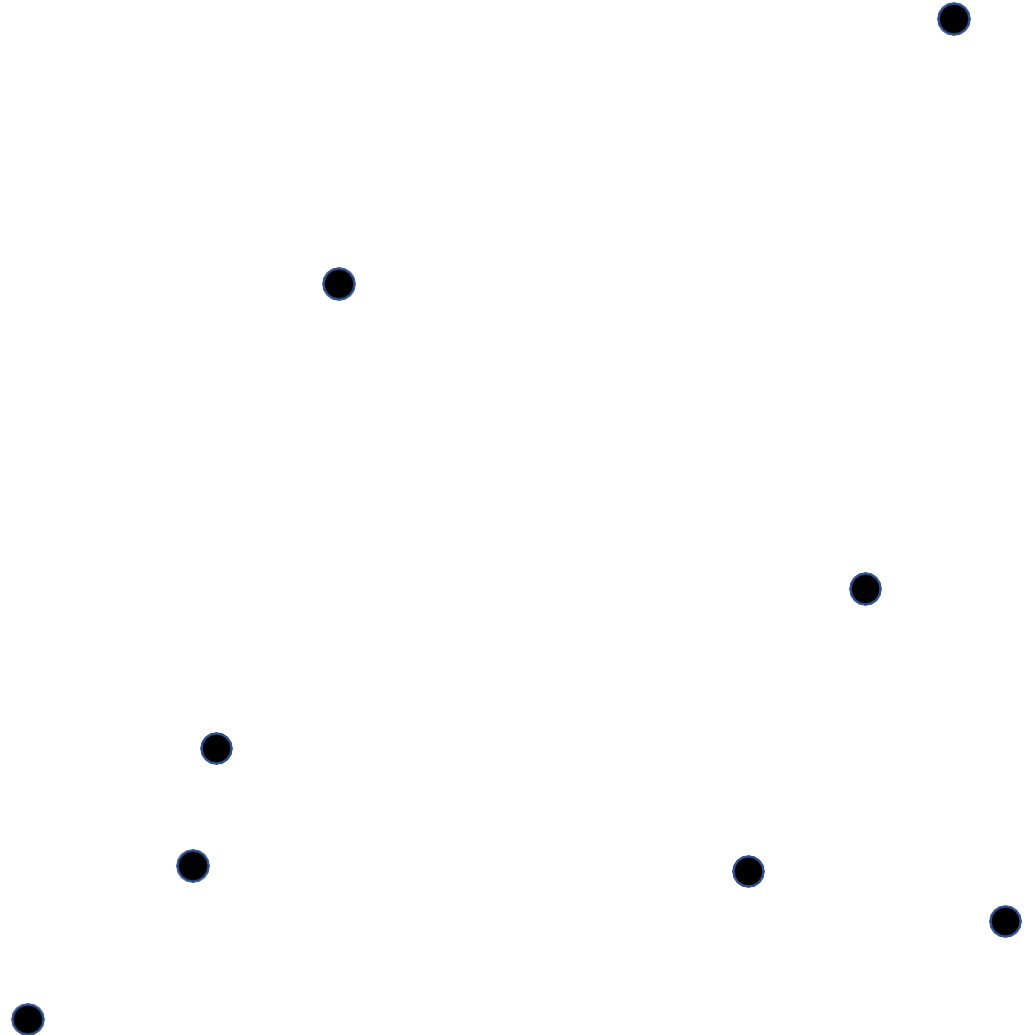
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... and divide that by Δt if speed is the desired metric.

! Although SLD is easy to quantify, it is also *heavily biased*.

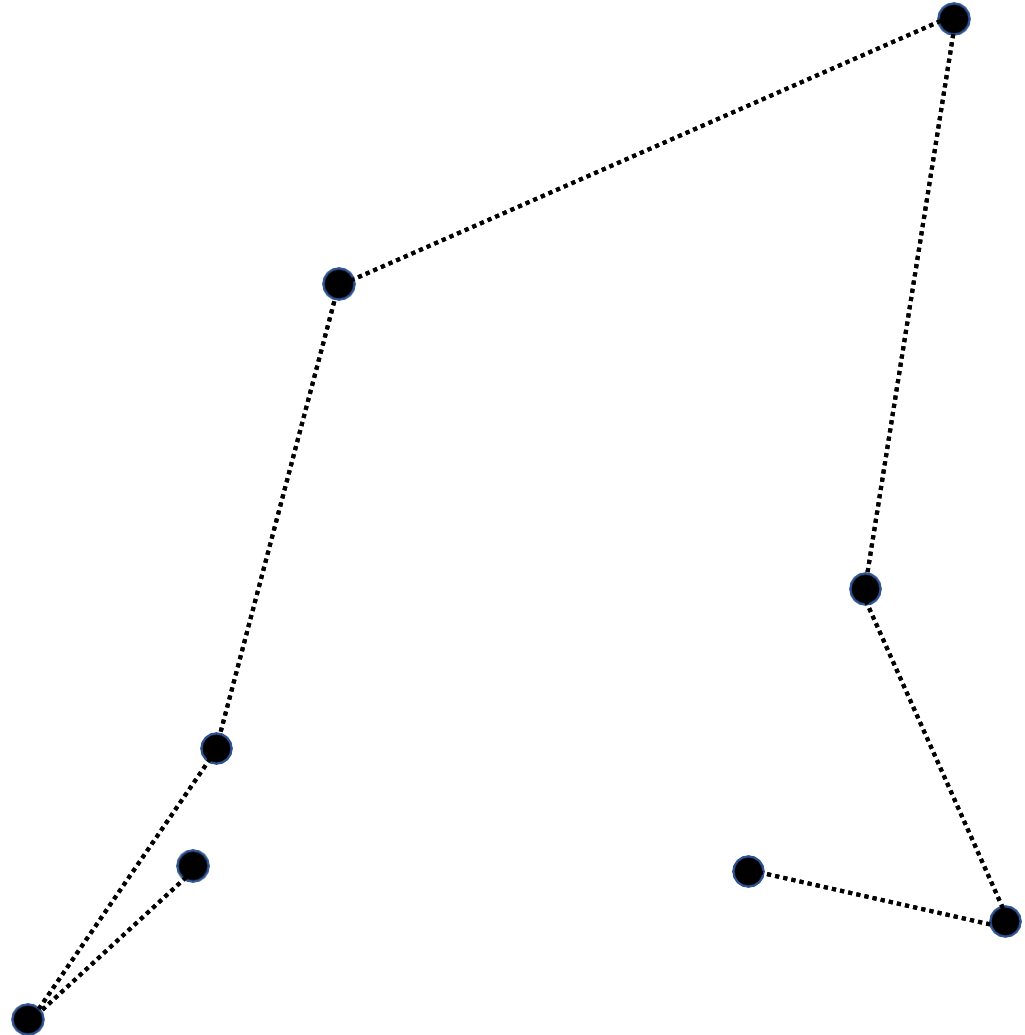
Estimating speed & distance

- Easy to quantify



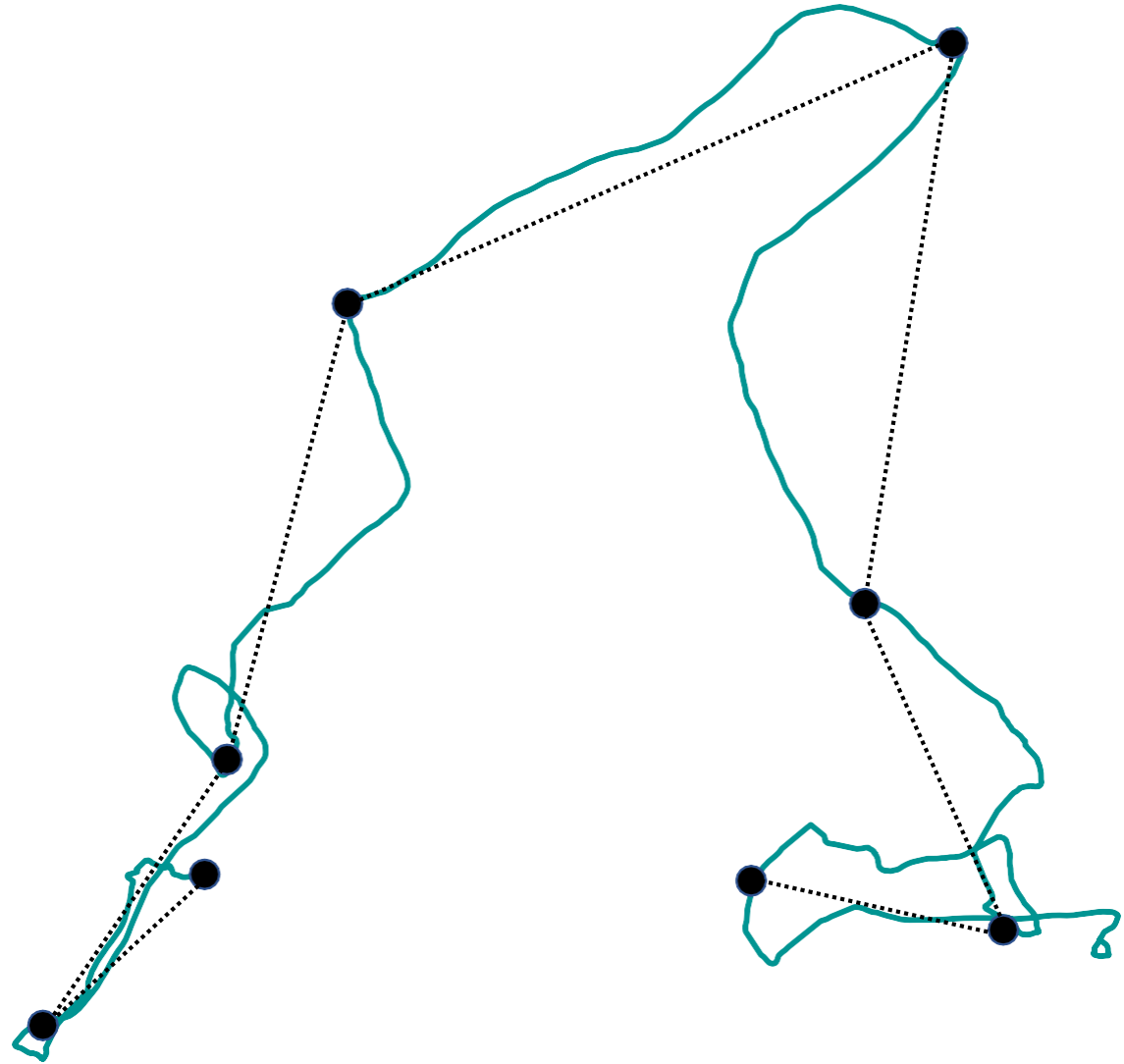
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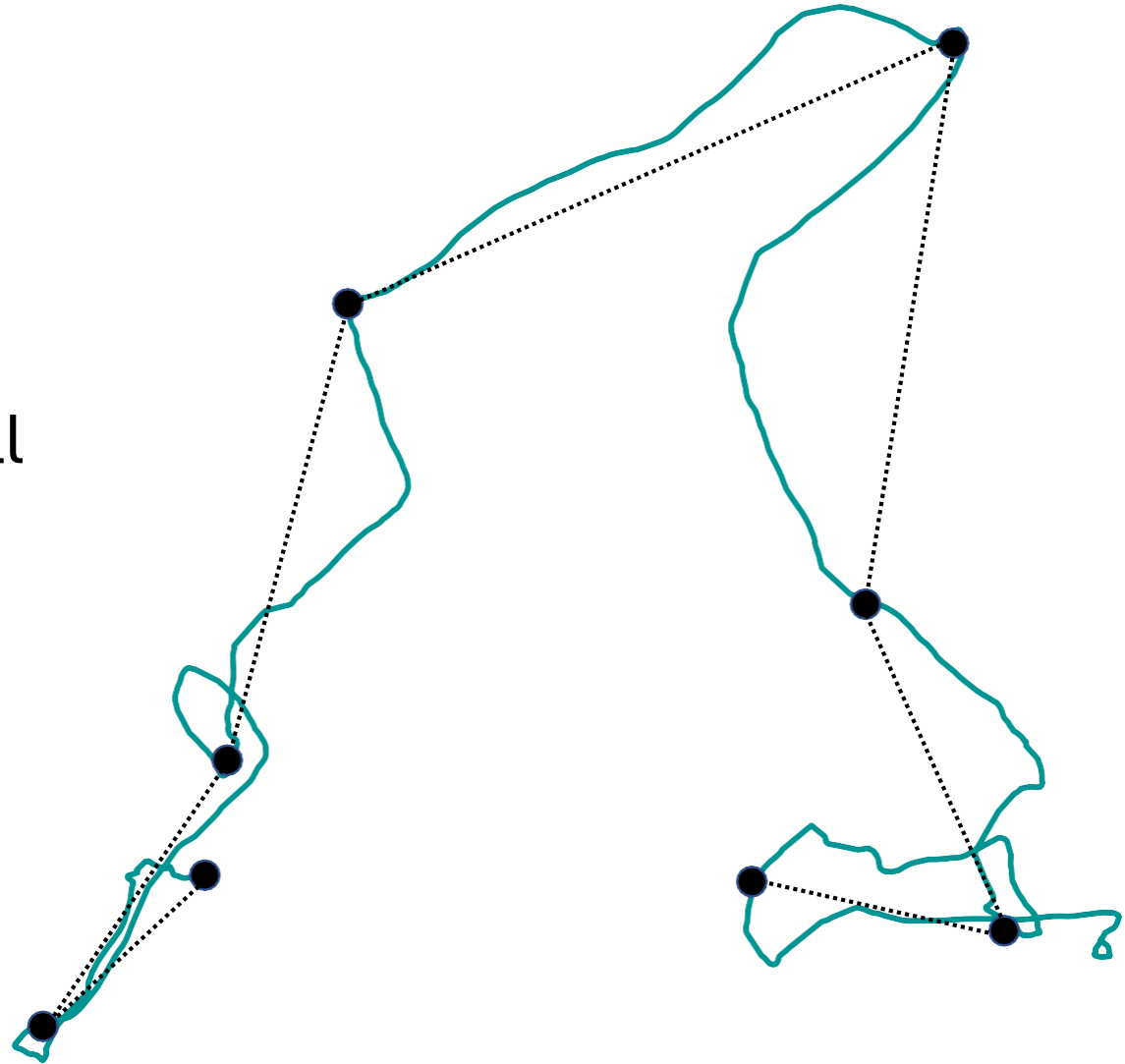
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- Unless the animal moved in a perfectly straight line, SLD will always **underestimate** the distance travelled



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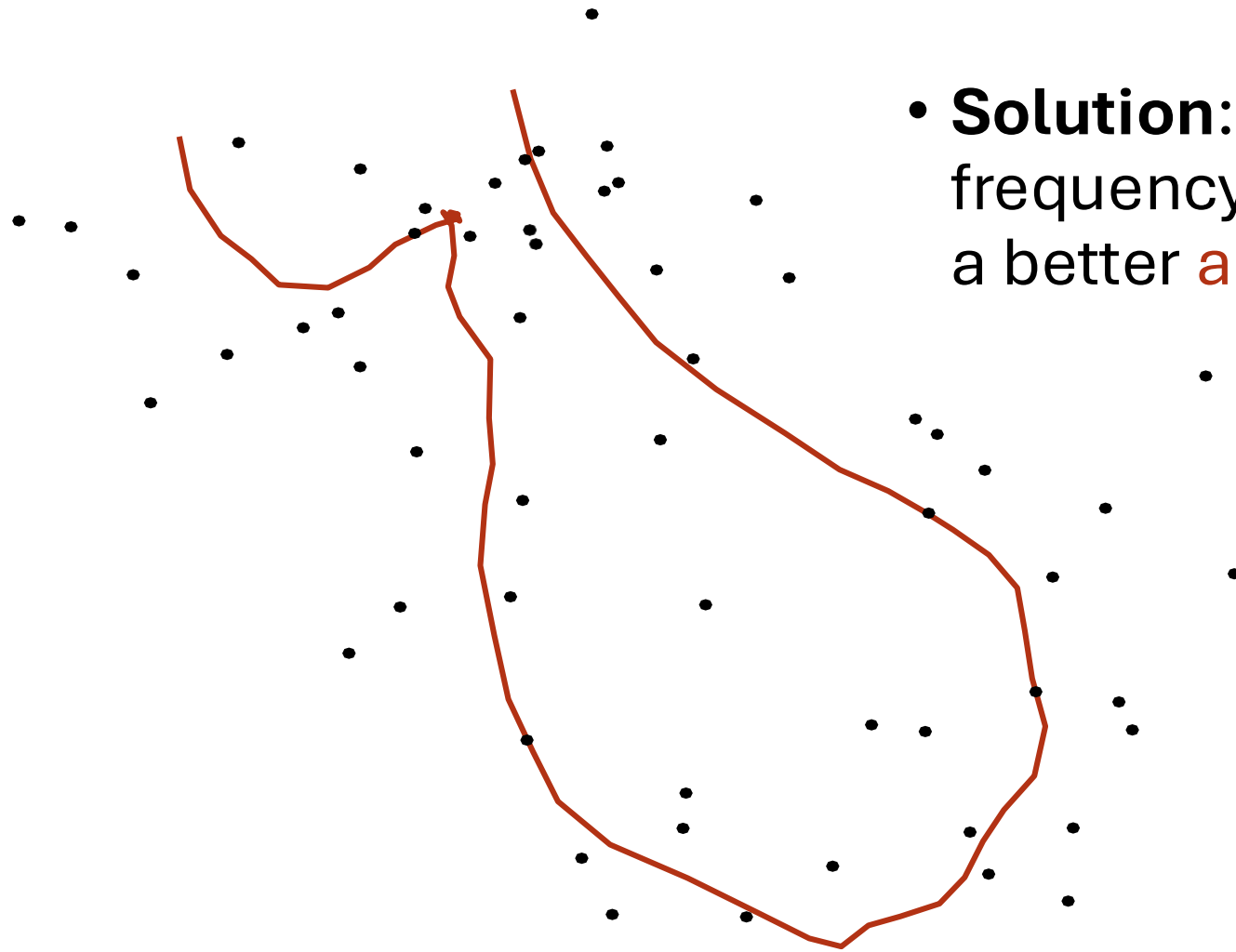
- Easy to quantify, but heavily biased
- Unless the animal moved in a perfectly straight line, SLD will always **underestimate** the distance travelled
- People have known this for decades
 - Bovet & Benhamou, 1988; Reynolds & Laundre, 1990; Rooney et al., 1998; Turchin, 1998; De Solla et al., 1999; Rowcliffe et al., 2012; Sennhenn-Reulen et al., 2017

Estimating speed & distance



- **Solution:** Increase the sampling frequency until the straight lines provide a better **approximation** of a curve?
 - Riemann sum

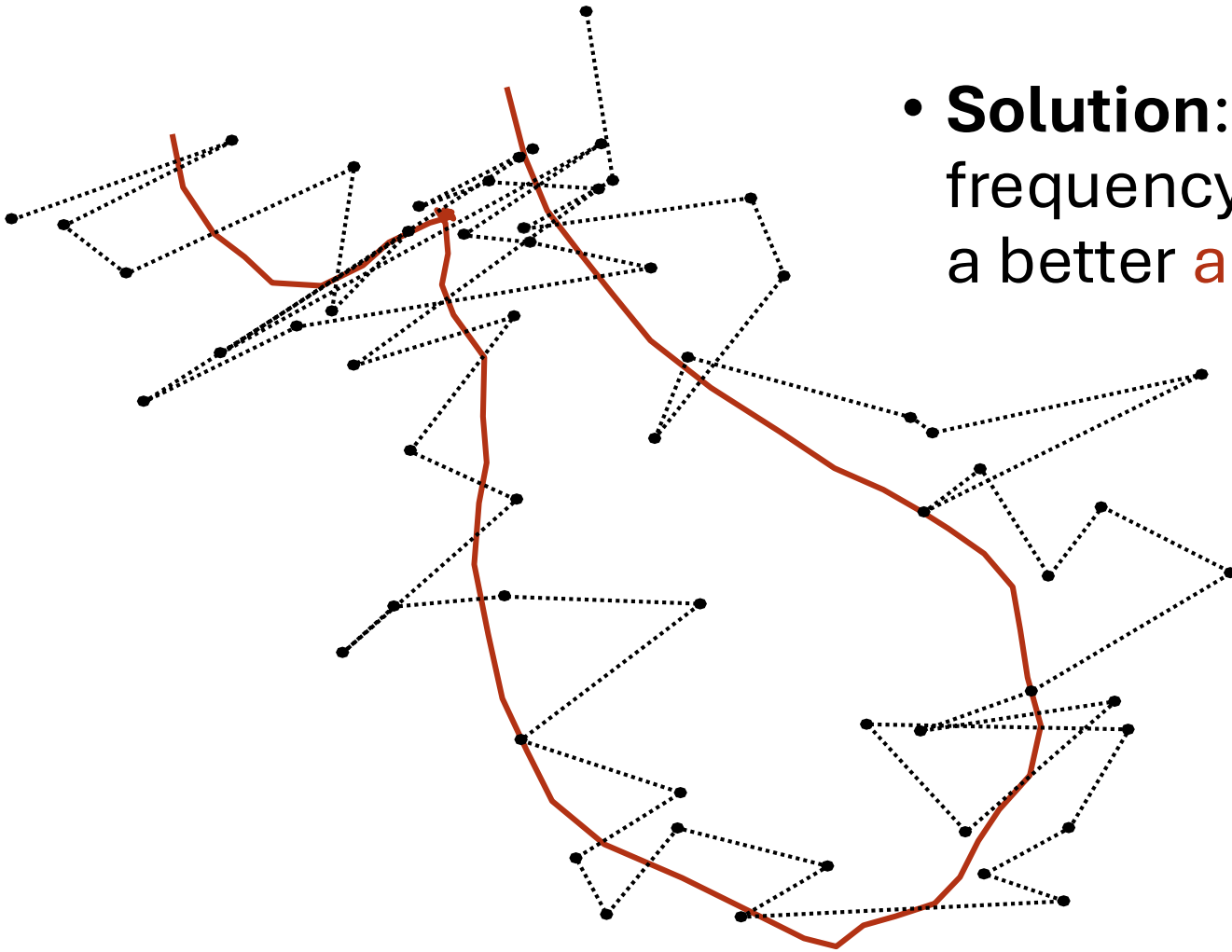
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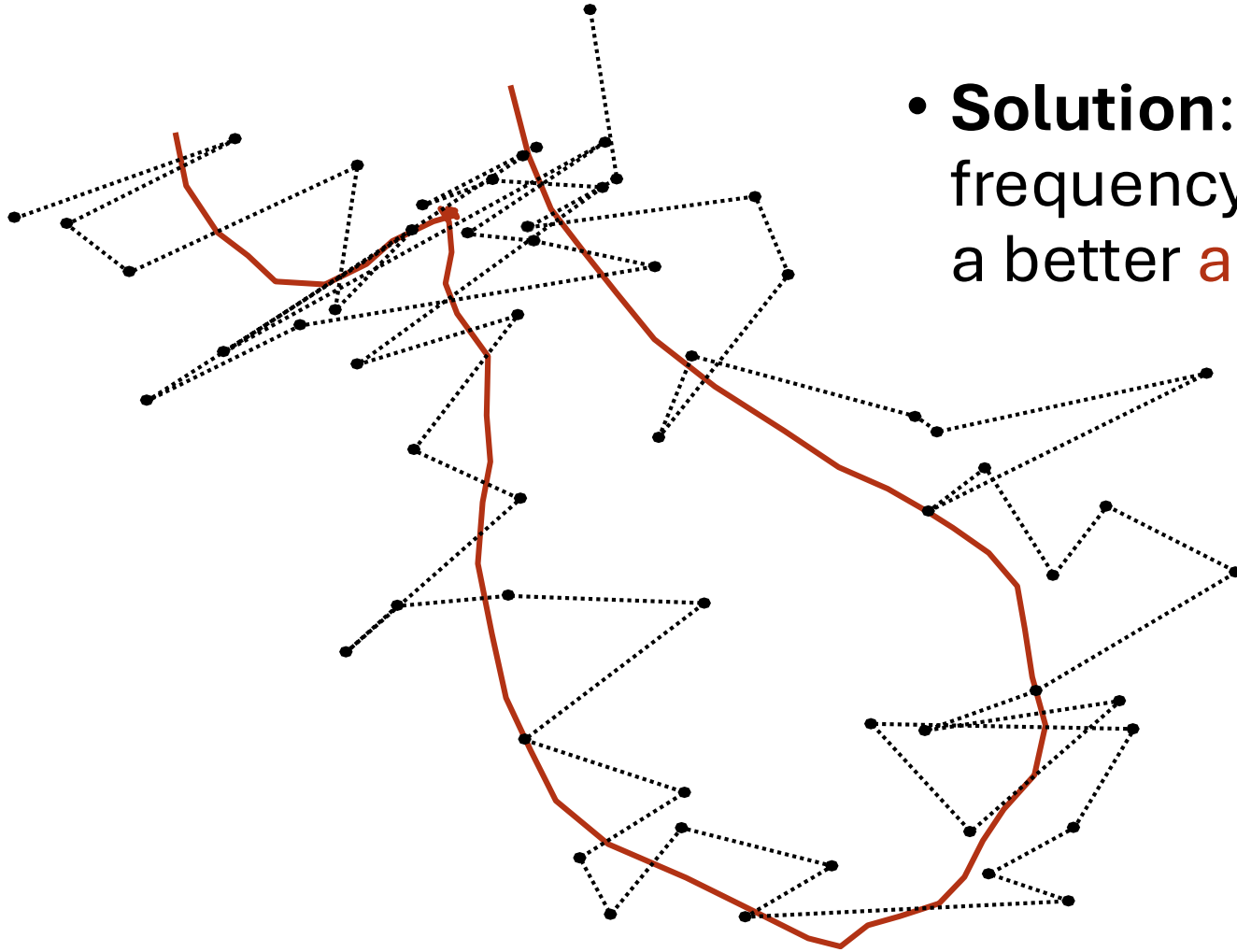
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However,

If error is uncorrelated in time, estimates diverge to **infinity** with infinite sampling frequency ($\Delta t \rightarrow 0$).

Why?

The actual distance traveled by the animal goes to 0 in the limit where $\Delta t \rightarrow 0$, but the magnitude of uncorrelated measurement error is **independent of Δt** .

Estimating speed & distance

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$$\lim_{\Delta t \rightarrow 0} \left| \frac{\Delta}{\Delta t} \underbrace{(\mathbf{r} + \mathbf{error})}_{\text{observable}} \right| = \infty$$

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- Few people have been thinking about this
 - Ranacher et al., 2015

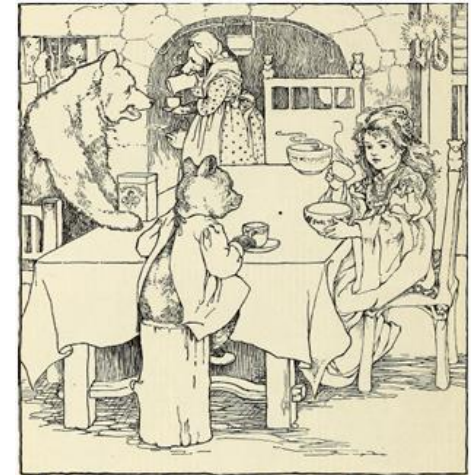
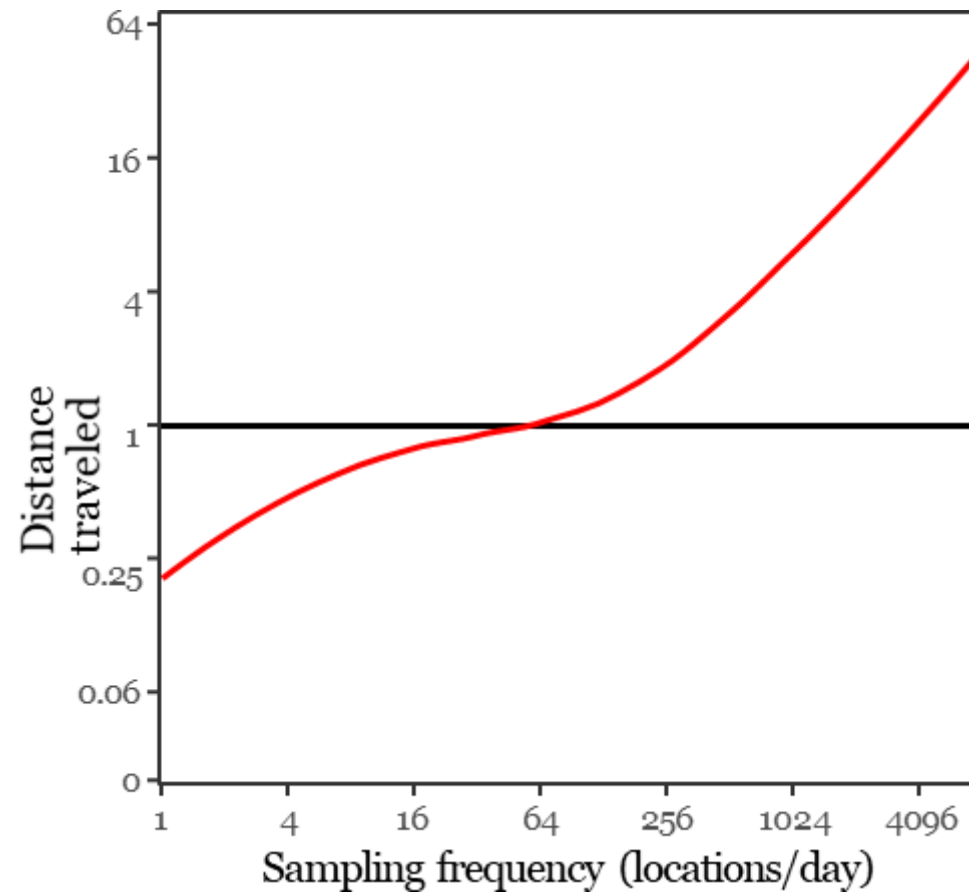
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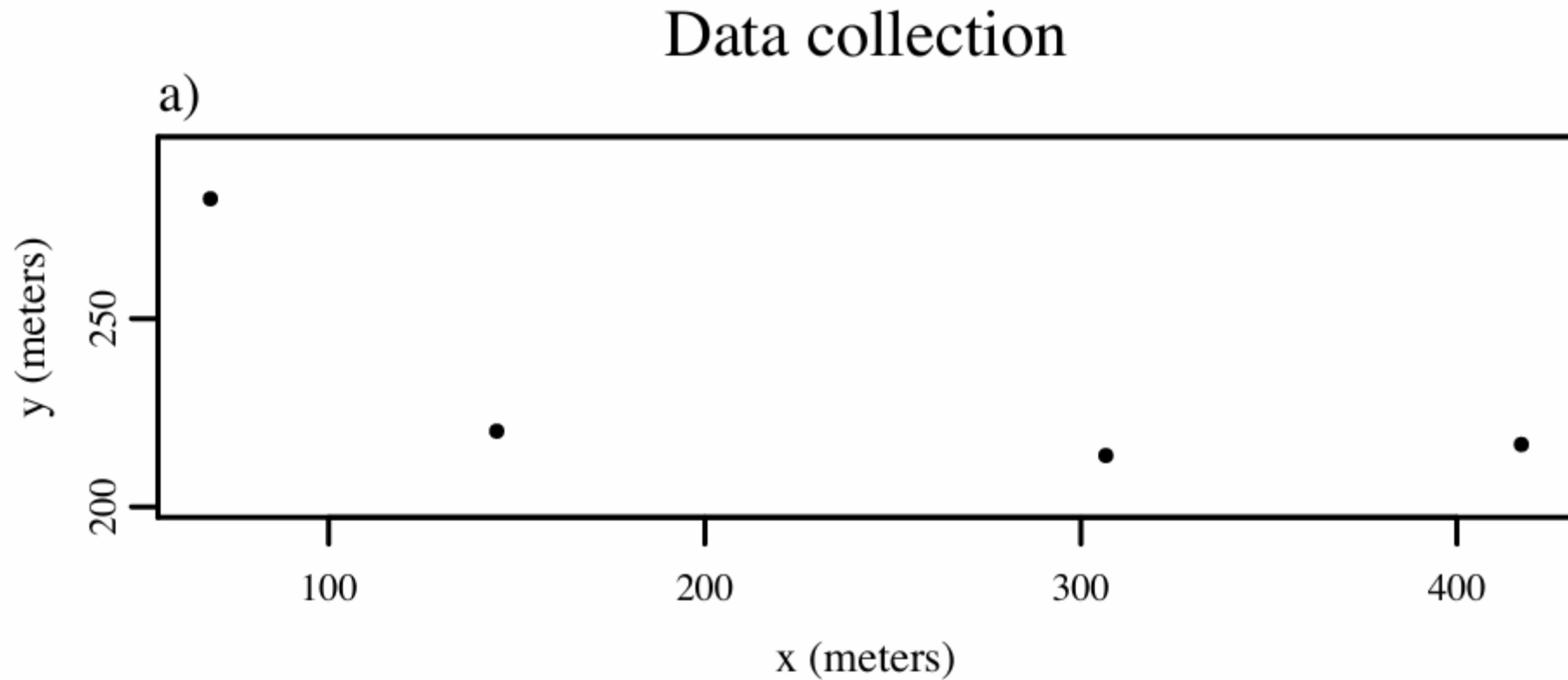
Estimating speed & distance



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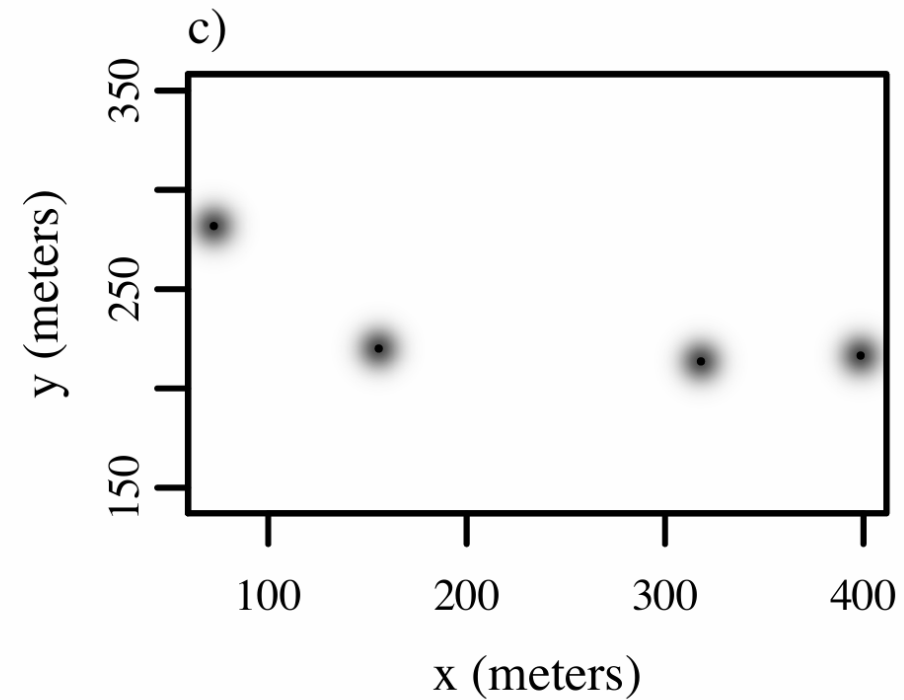
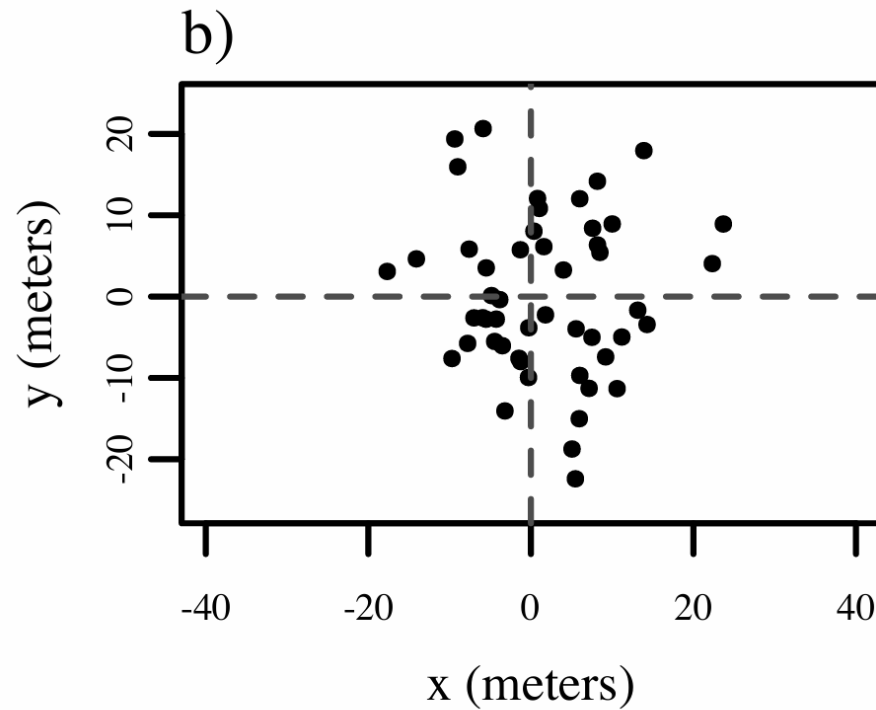
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How does it work?



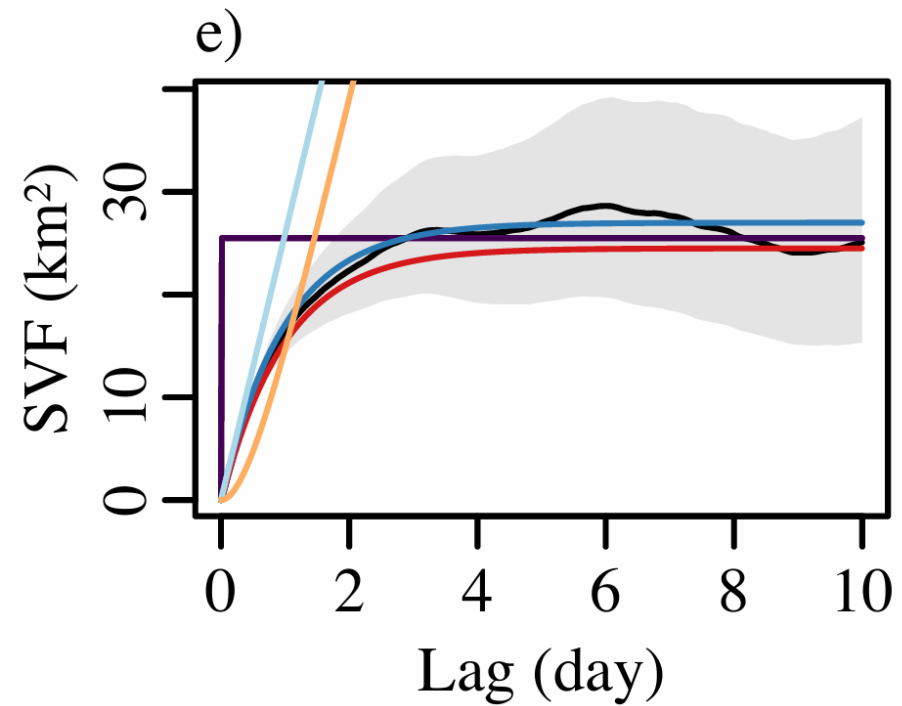
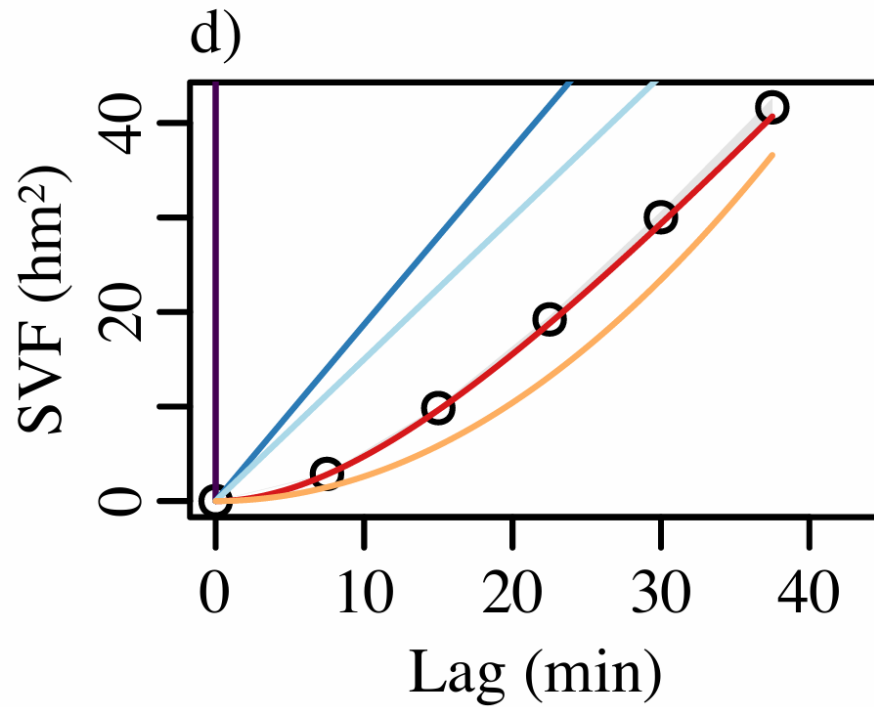
How does it work?

Error calibration



How does it work?

Model selection



How does it work?

- **Continuous-time speed and distance (CTSD)**
 - Noonan et al. (2020)

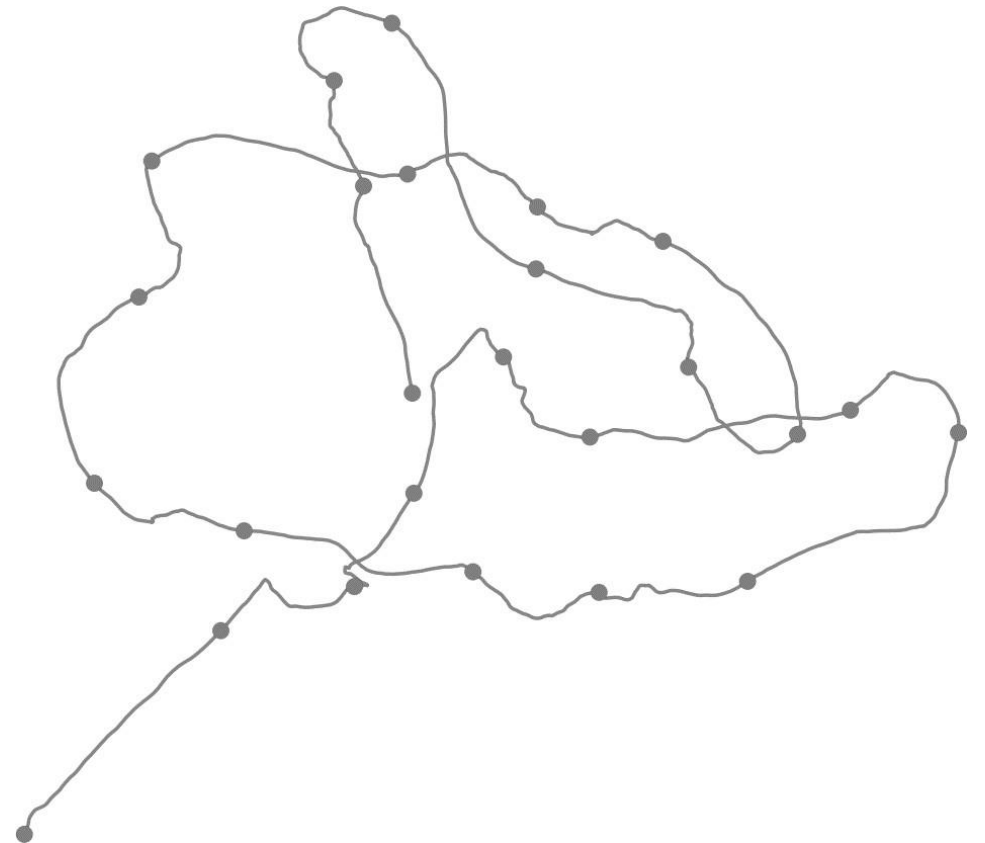
Model	Autocorrelation			Parameters:
	Position	Velocity	Restricted	
IID	No	No	Yes	$\tau = \text{NULL}$
BM	Yes	No	No	$\tau = \infty$
OU	Yes	No	Yes	$\tau = \tau_p$
IOU	Yes	Yes	No	$\tau = \{\infty, \tau_v\}$
OUF	Yes	Yes	Yes	$\tau = \{\tau_p, \tau_v\}$



Requires a *correlated velocity model*.

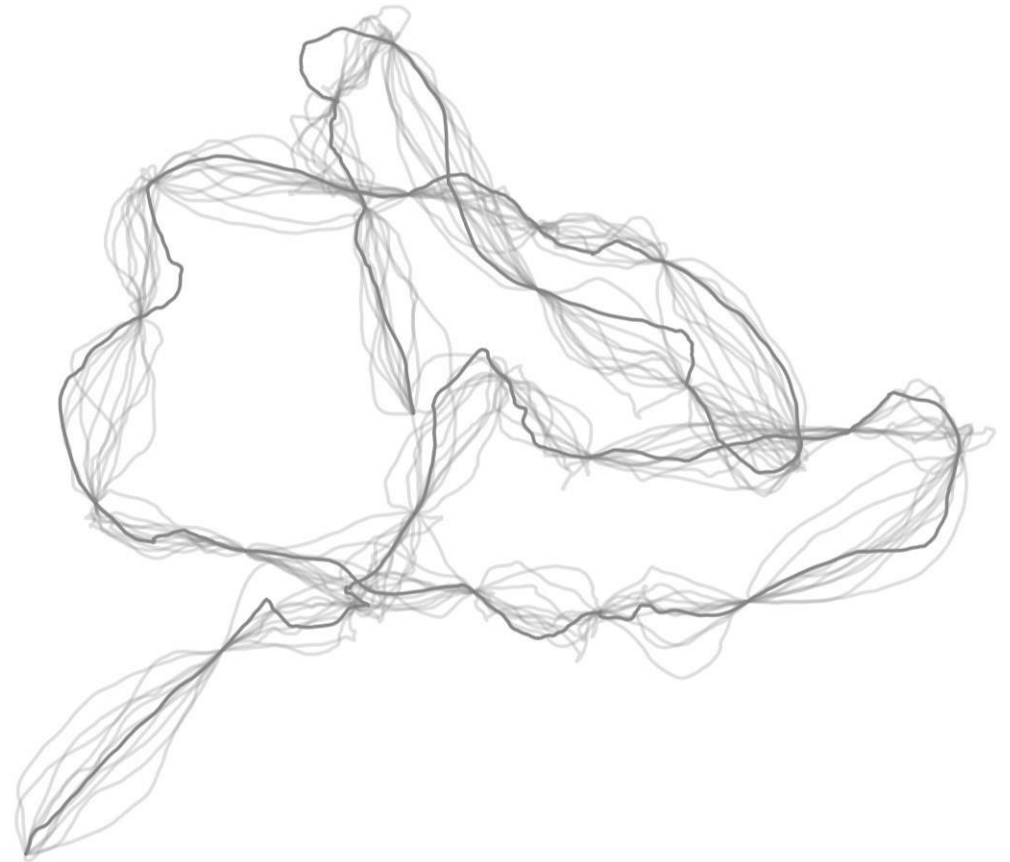
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- **Continuous-time speed and distance (CTSD)**
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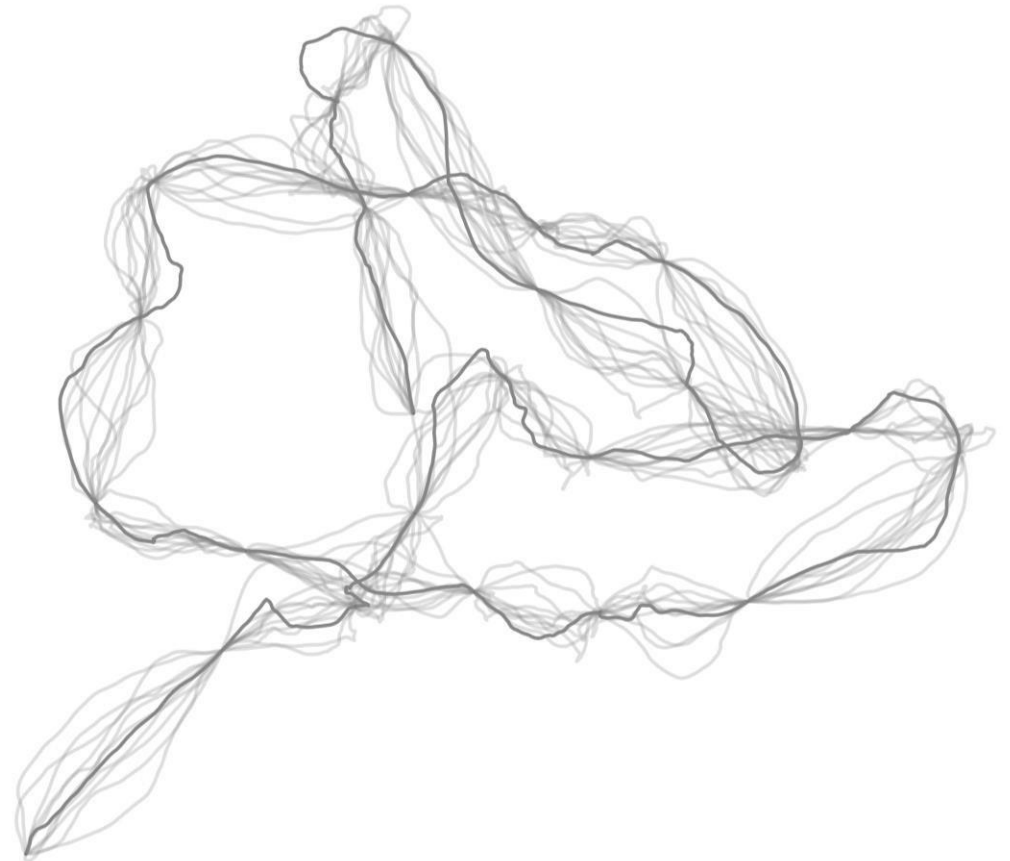


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- Until we get a **point estimate** (mean), and variance around the estimate (for the **confidence intervals**)



Estimating speed & distance

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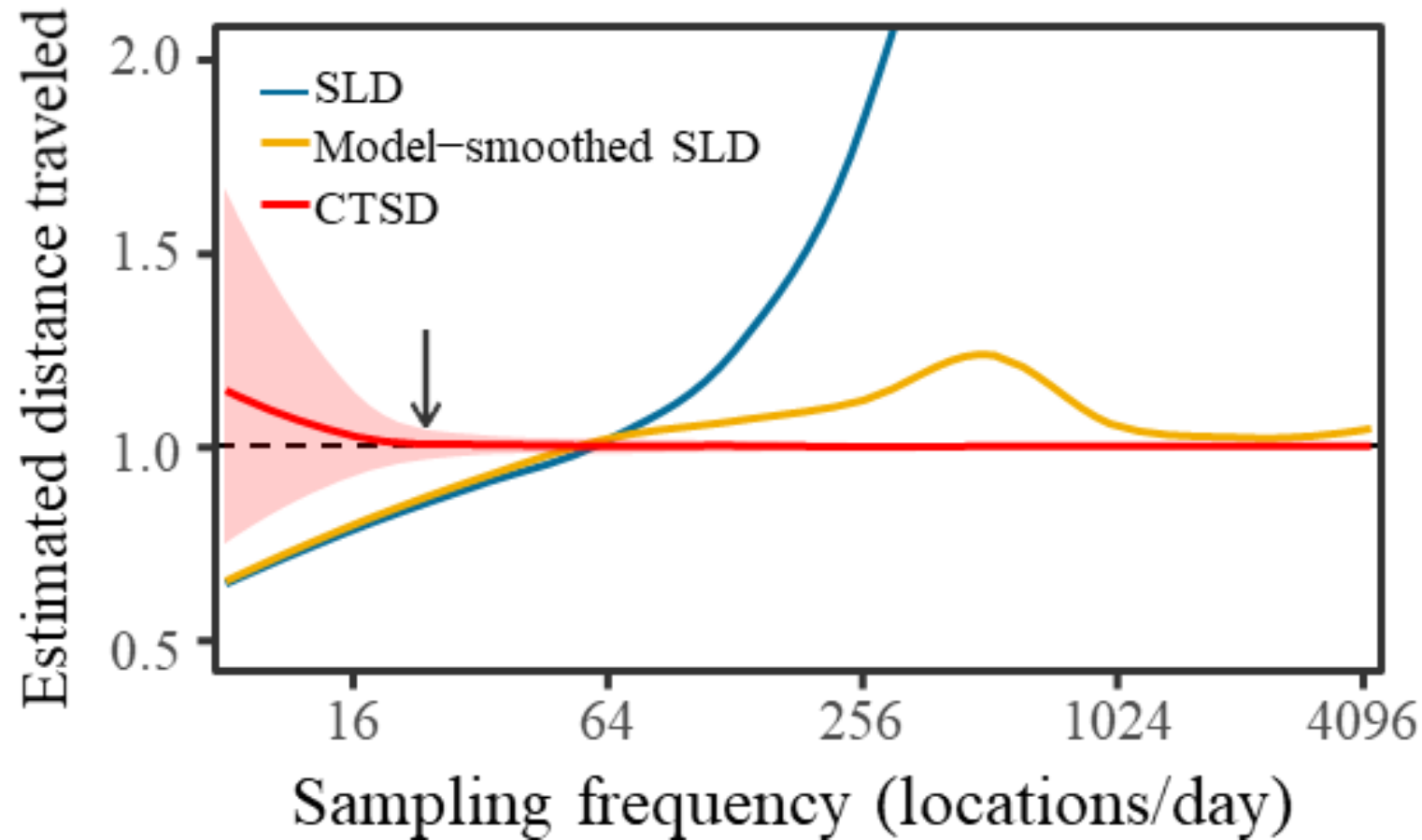
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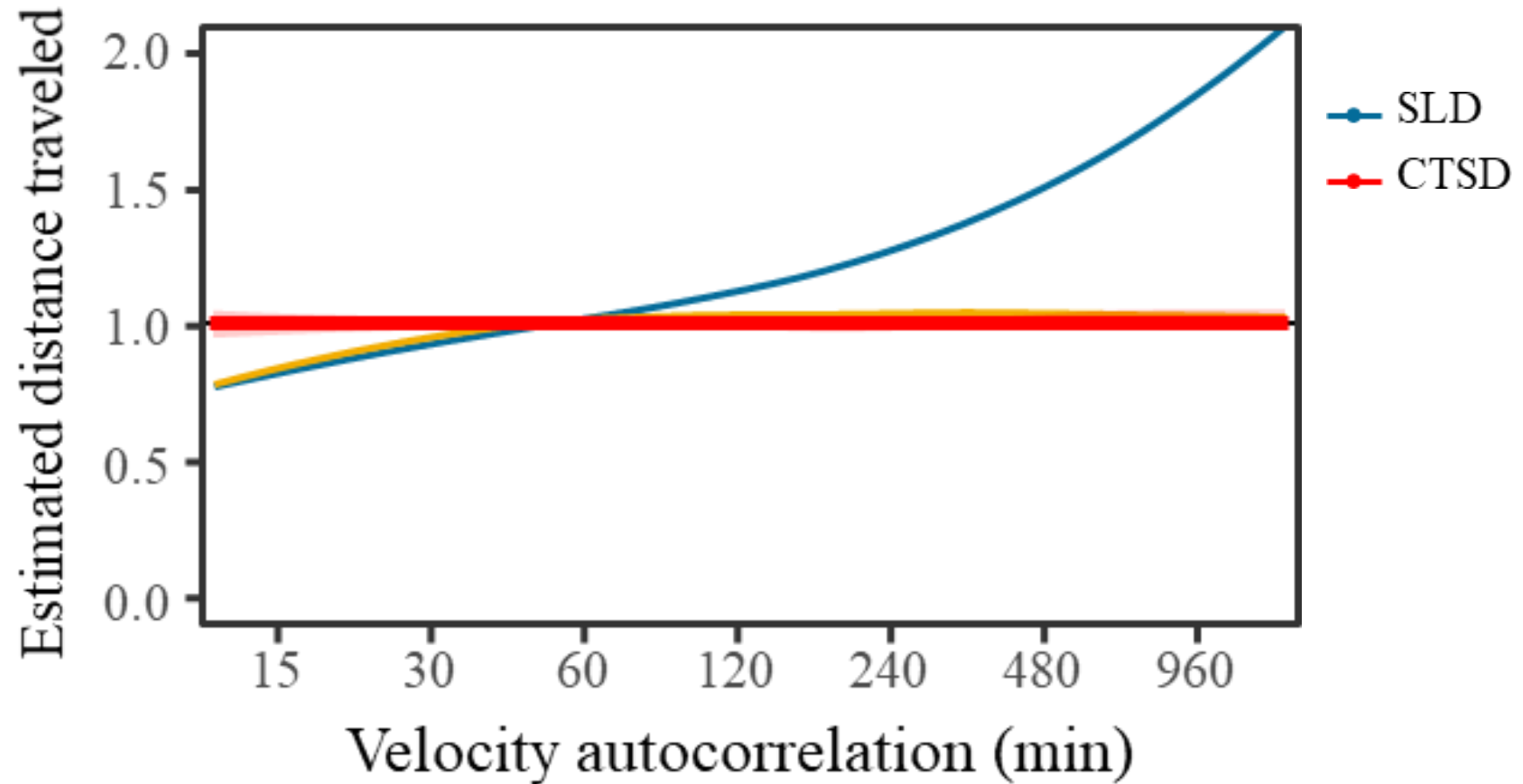
$$\text{DOF}_{\text{speed}} = \frac{\langle \bar{v} \rangle^2}{2 \text{VAR}[\hat{v}]}$$

$$\text{DOF}_{\text{distance}} = \frac{\langle d \rangle^2}{2 \text{VAR}[\hat{d}]}$$

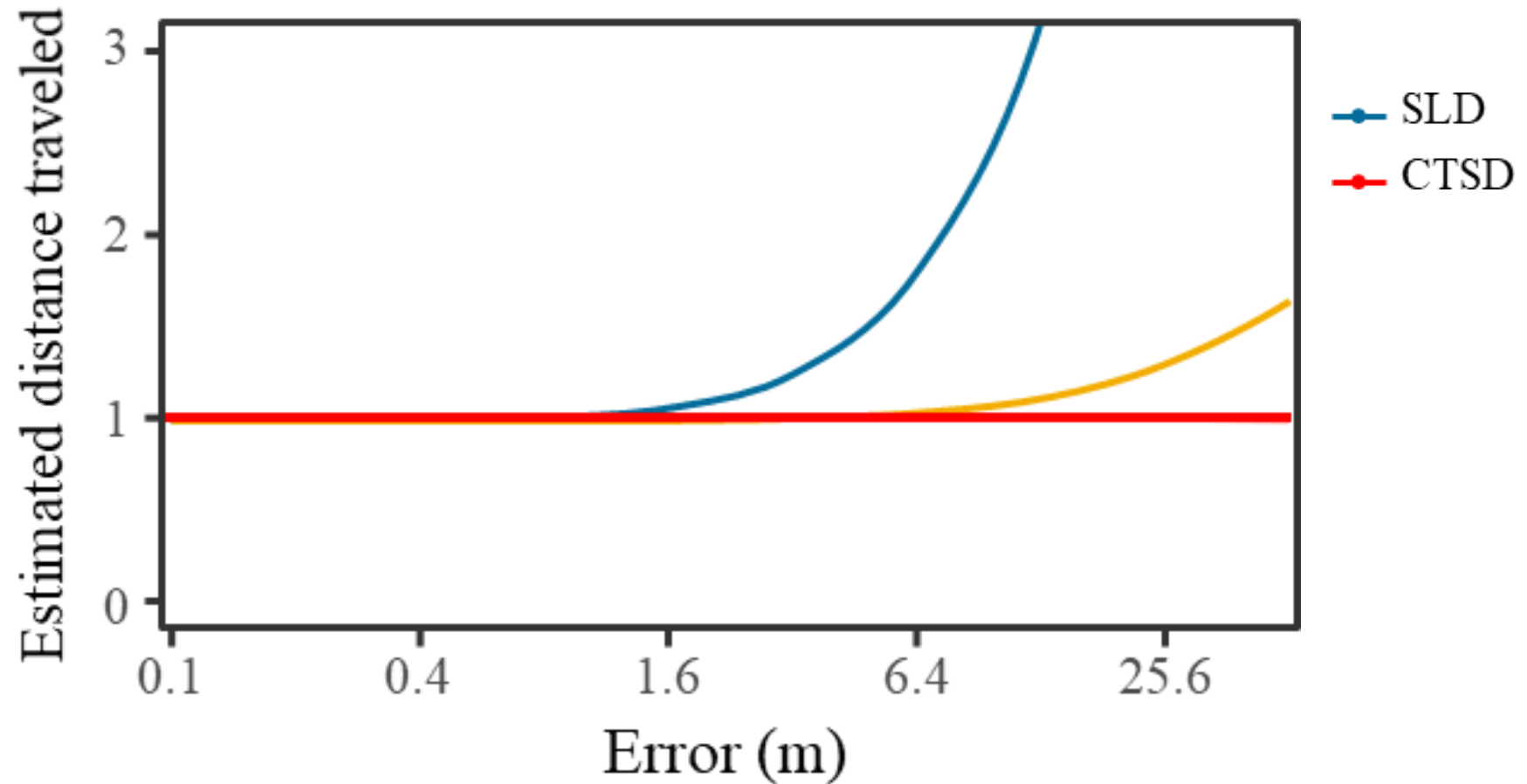
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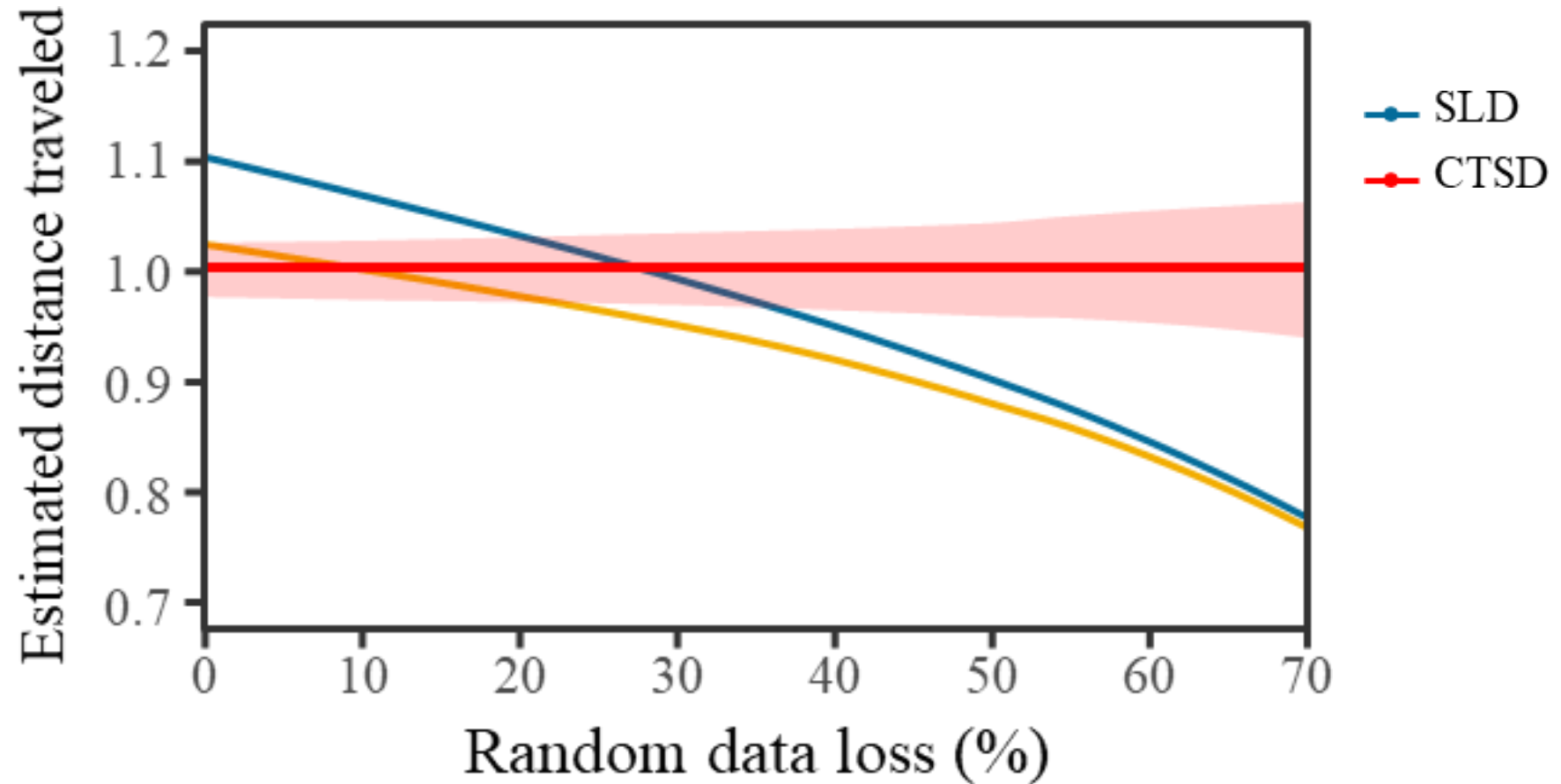
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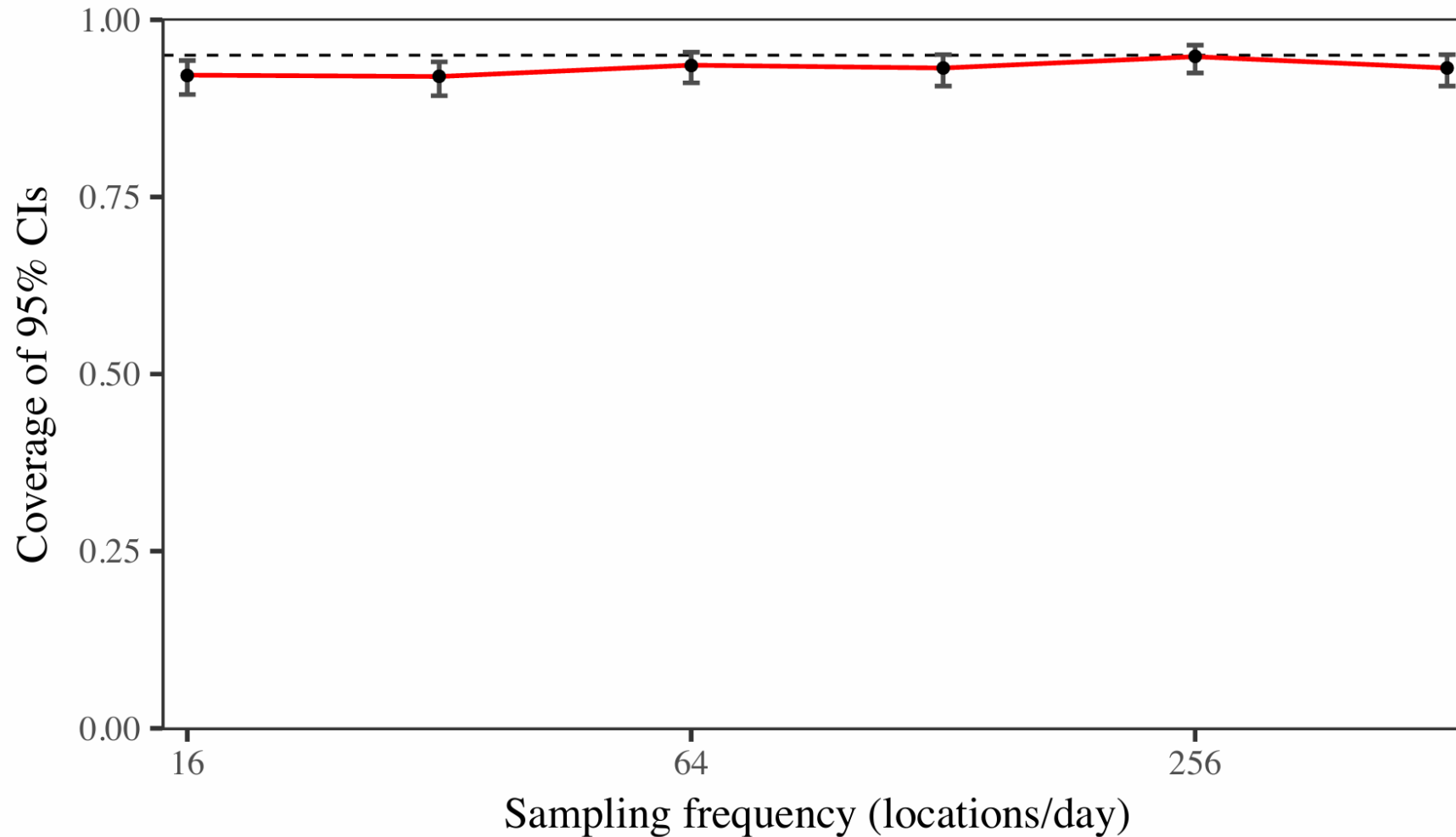
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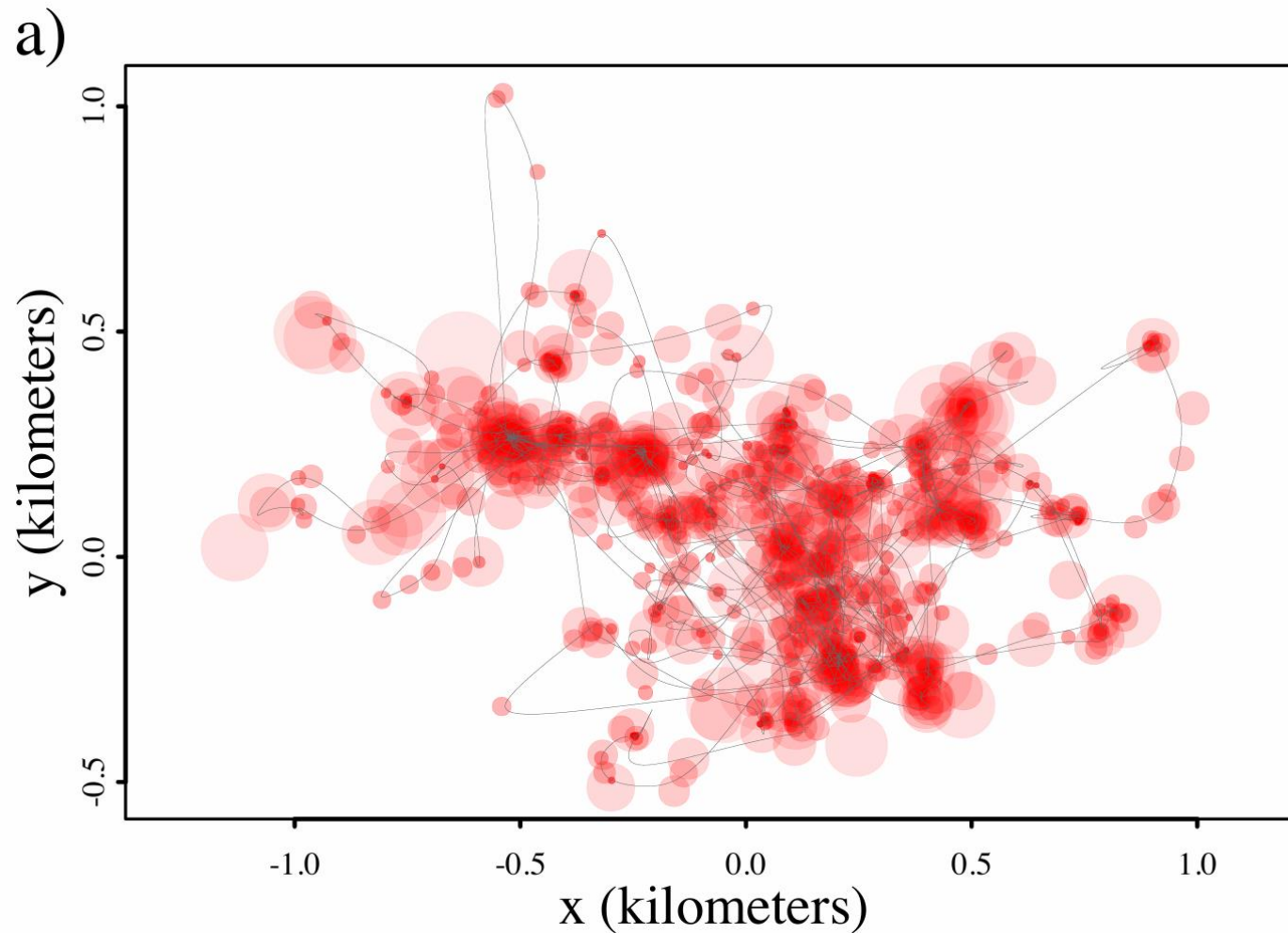
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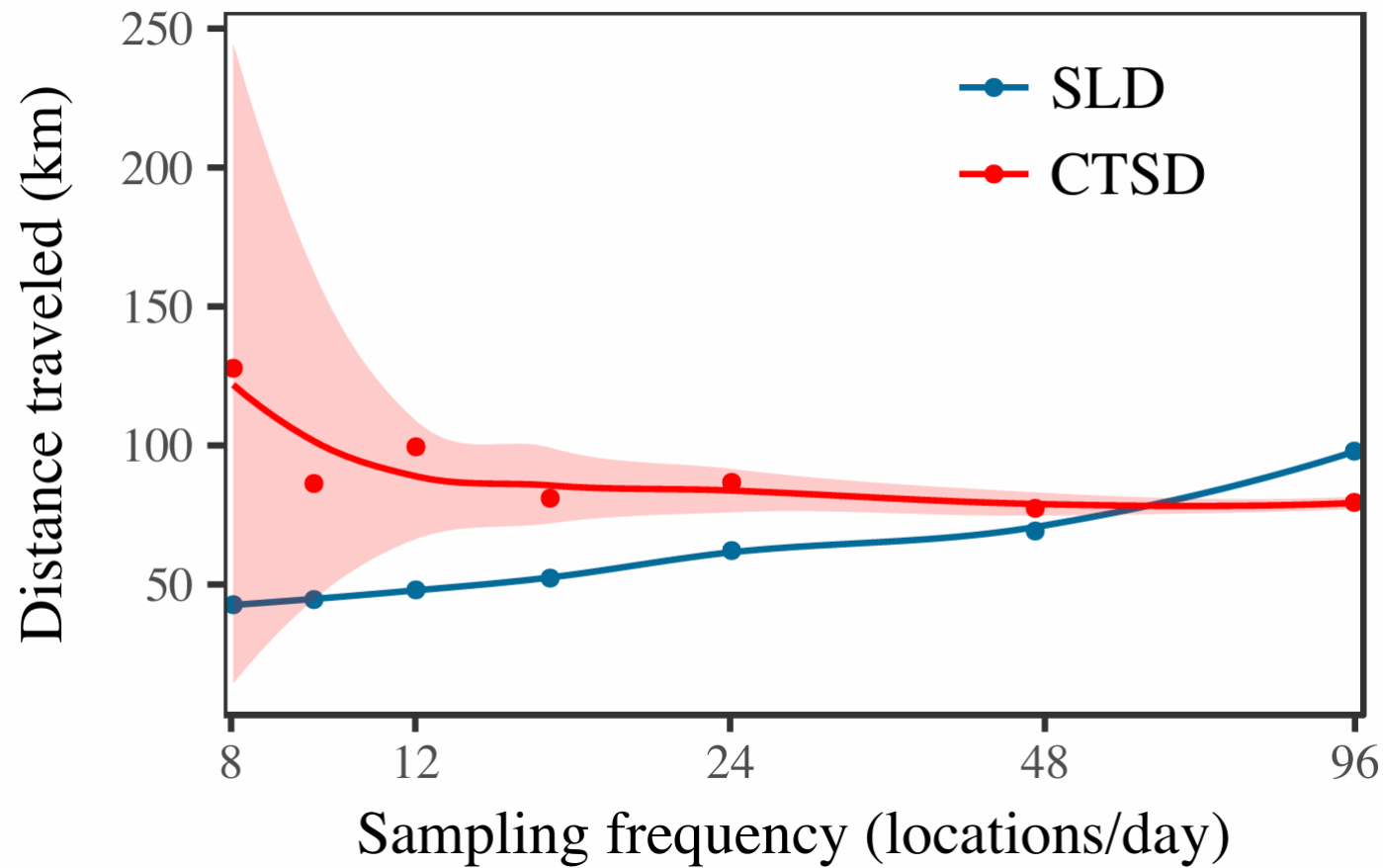
Applications – Mean speed



White-nosed coati
(*Nasua narica*)

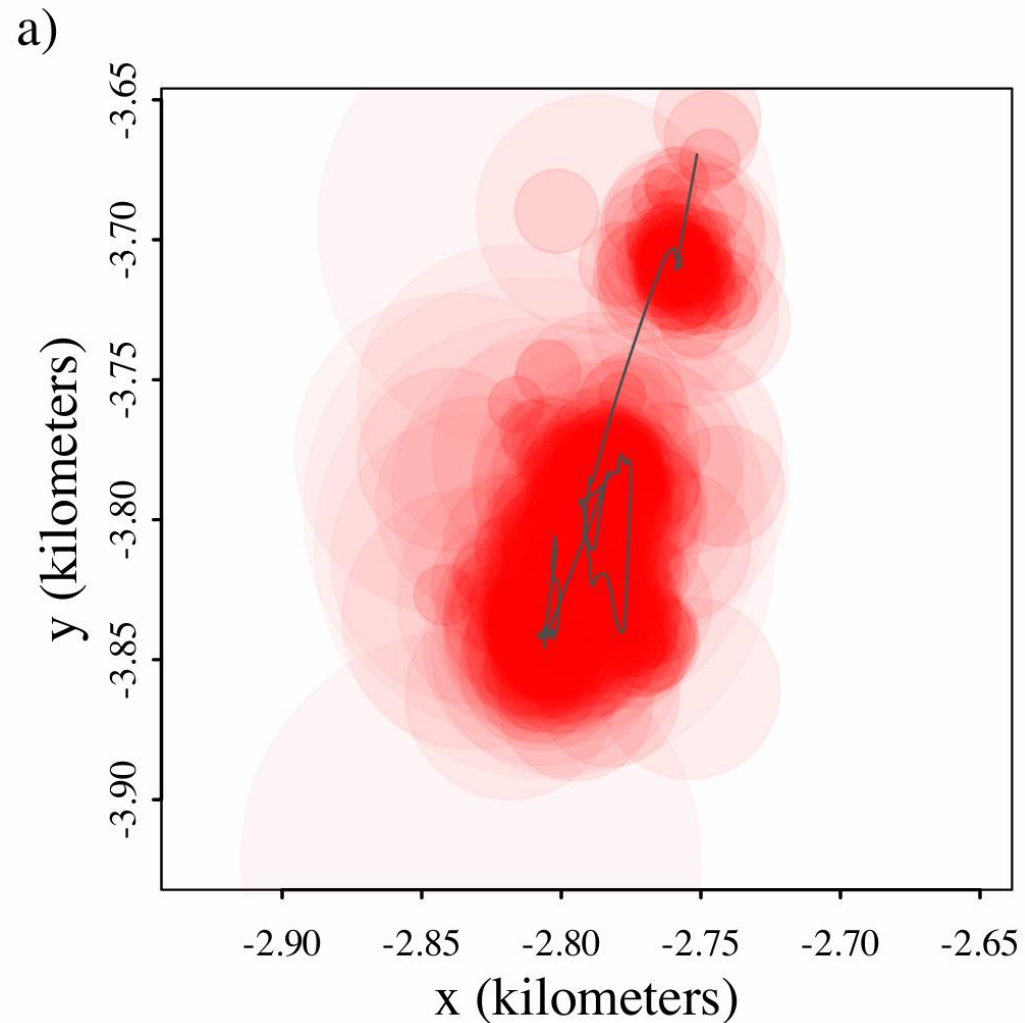
Applications – Mean speed

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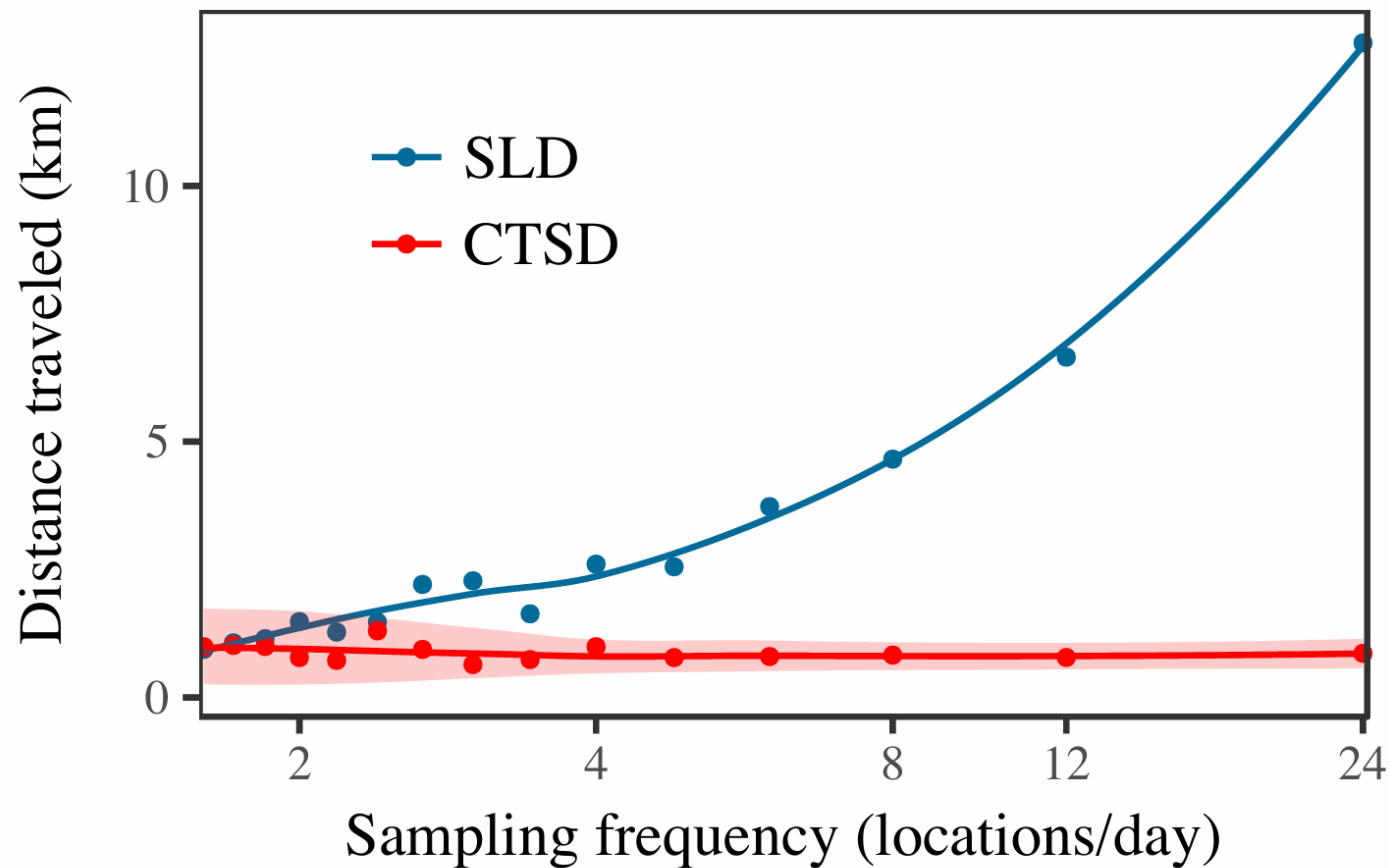


Wood turtle
(*Glyptemys insculpta*)



Applications – Mean speed

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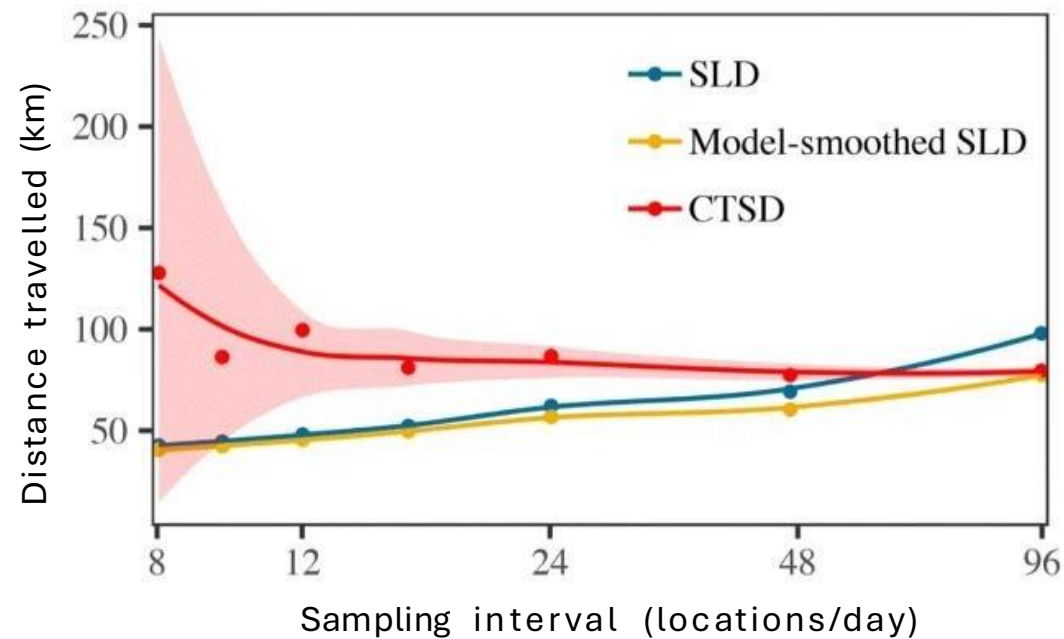
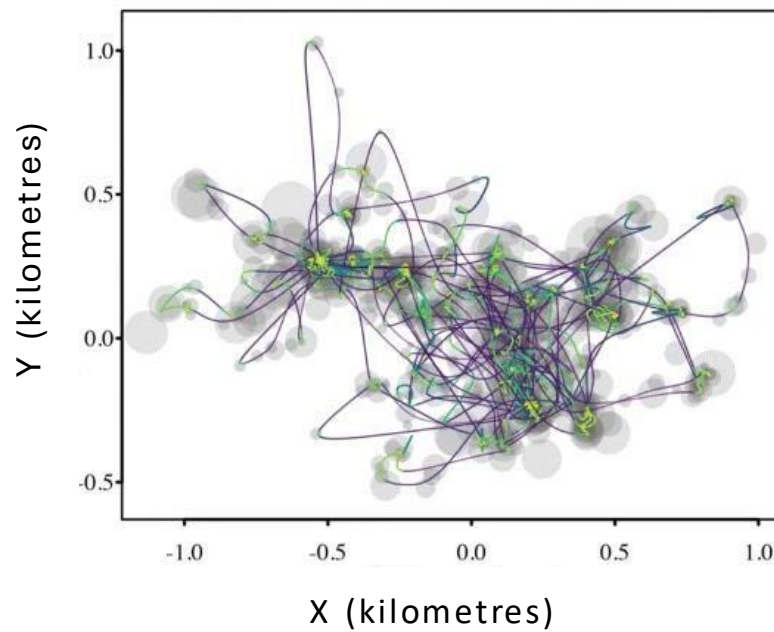
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Applications – Instantaneous speeds

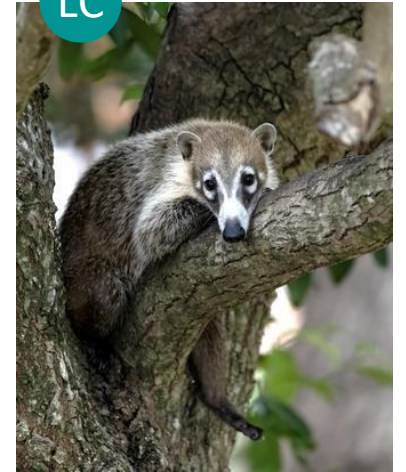
Sampling frequency (max)

1 fix every 15 minutes



LC

Chuck Homler

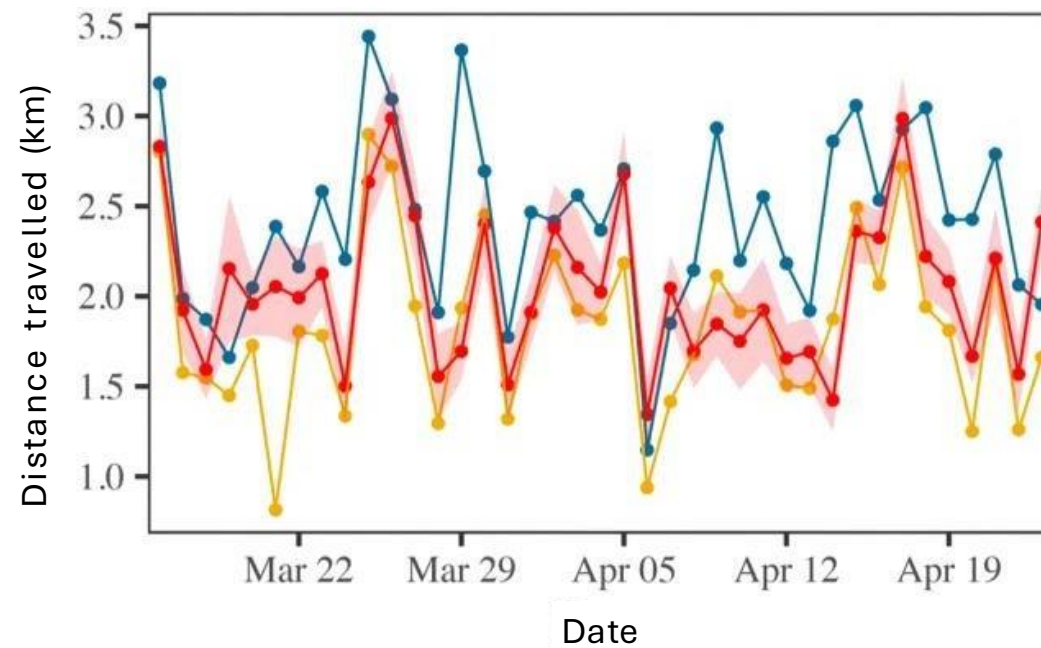
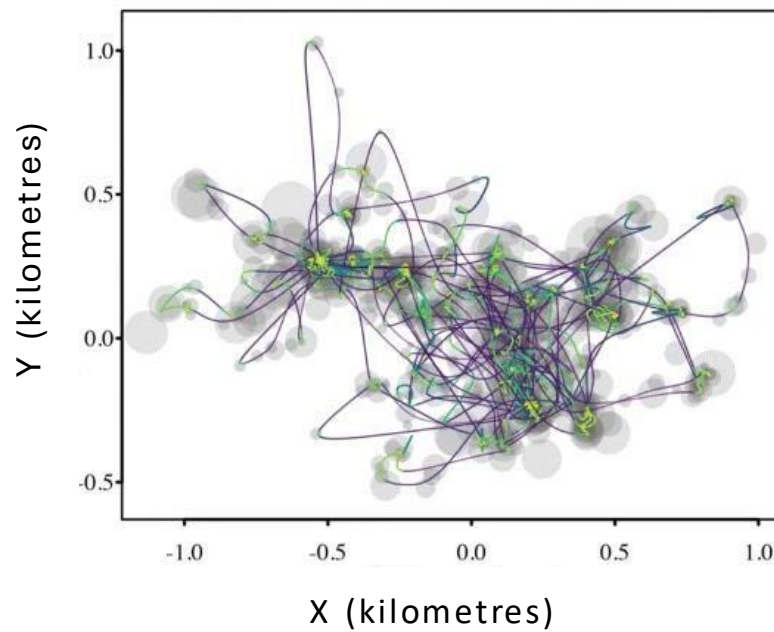


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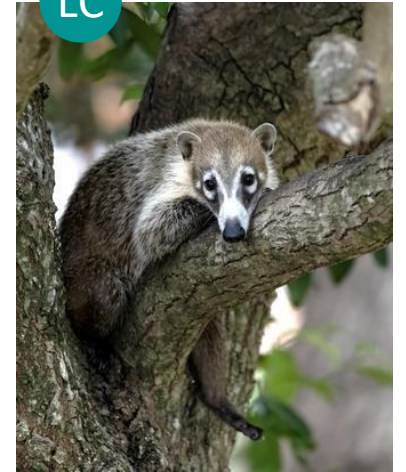
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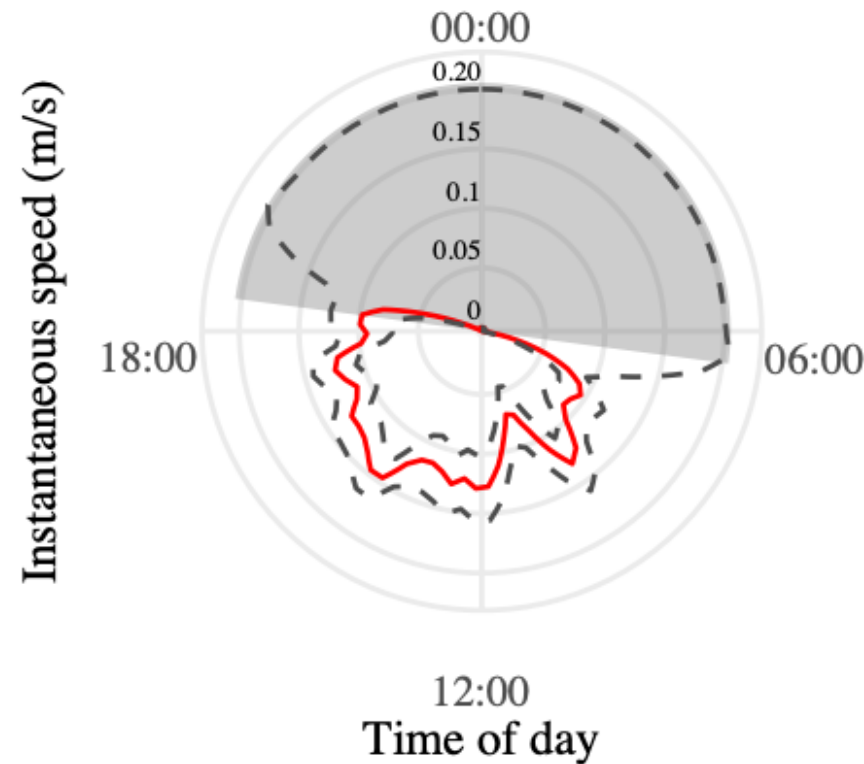
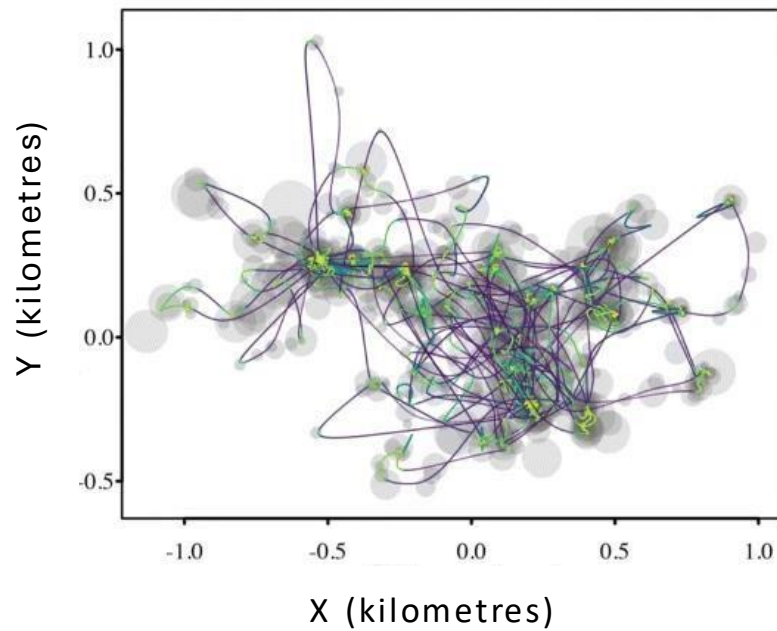
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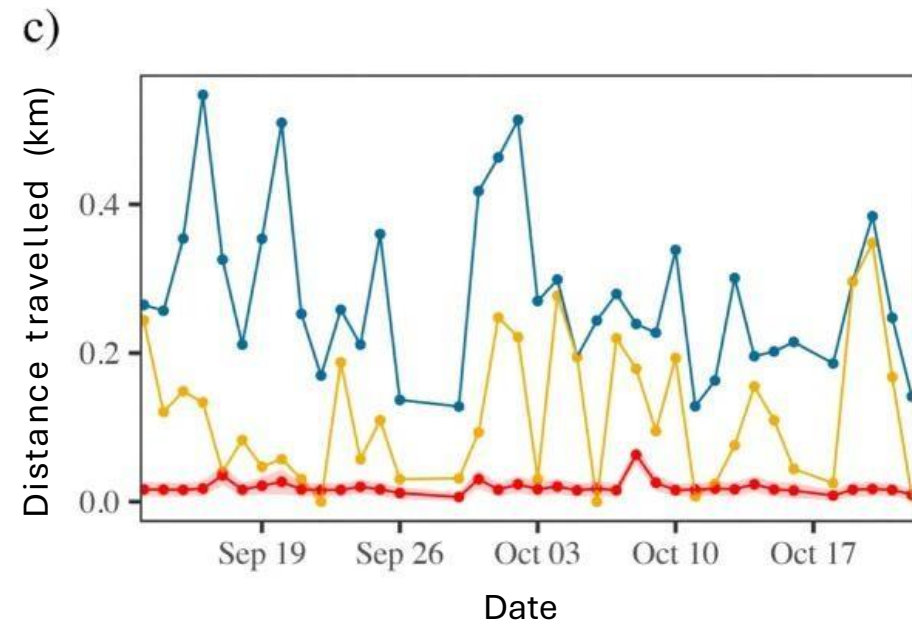
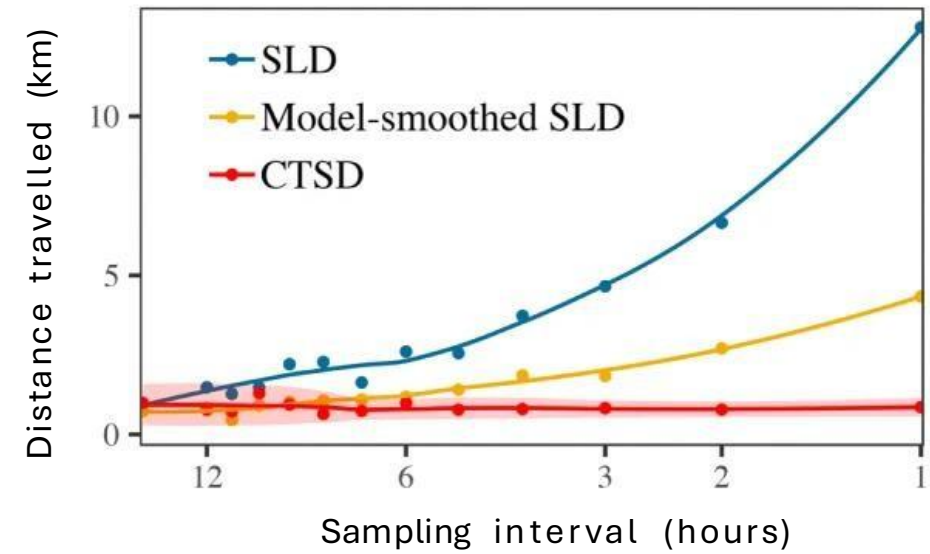
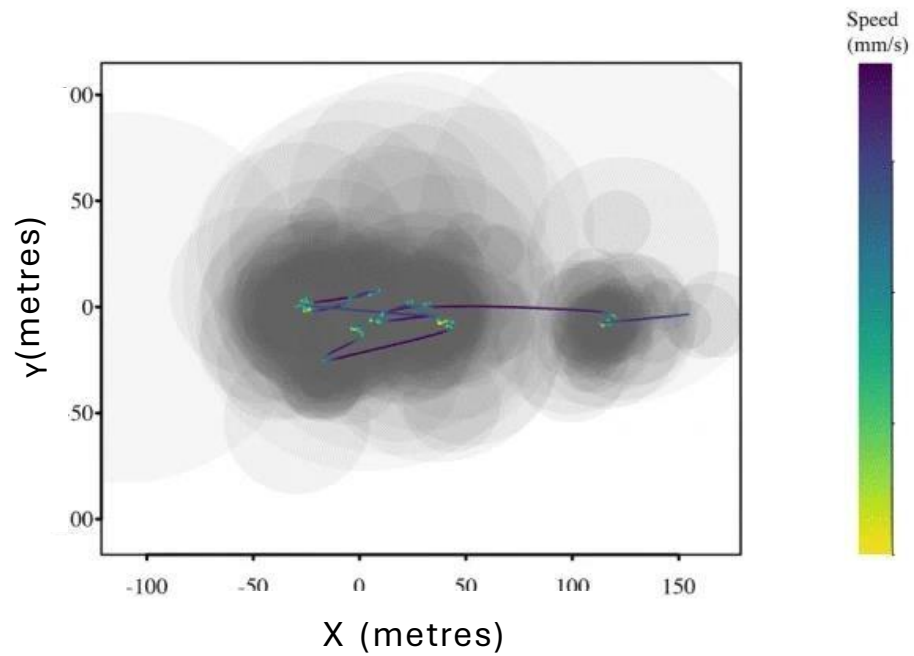
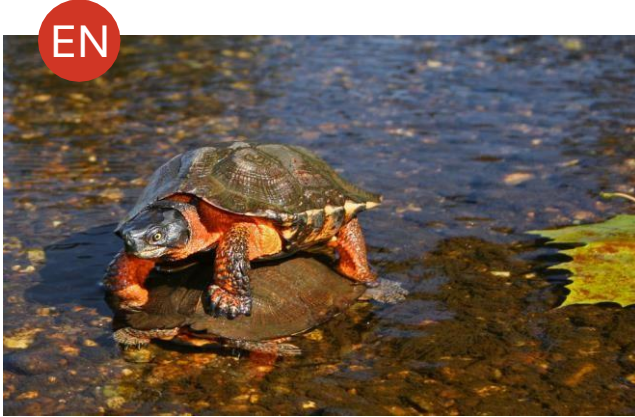
Applications – Instantaneous speeds

- Describing behaviors



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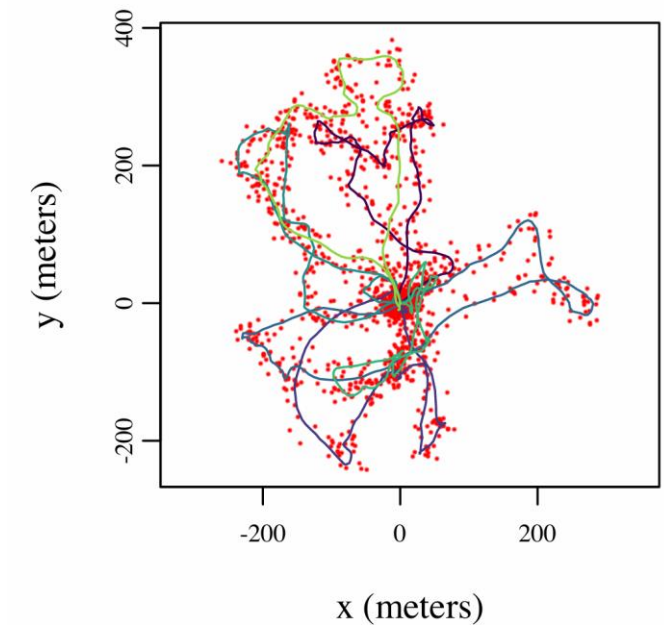
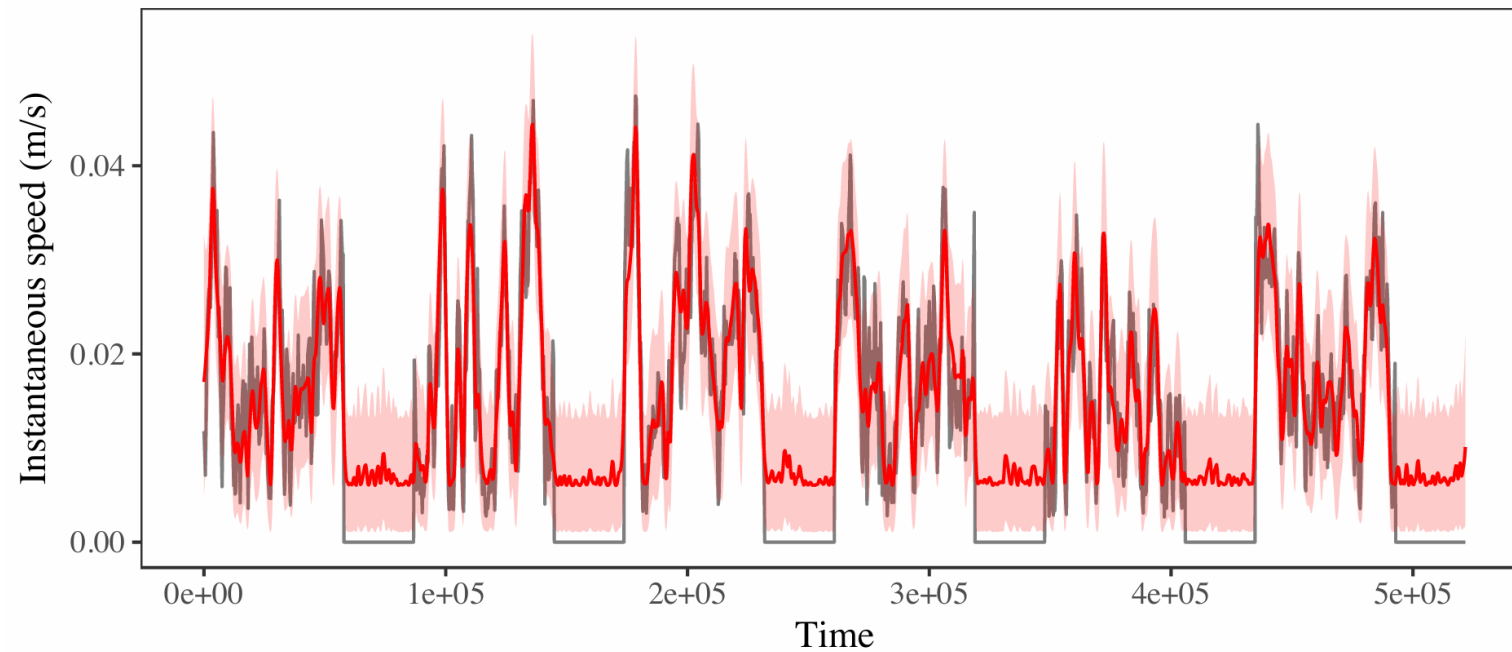
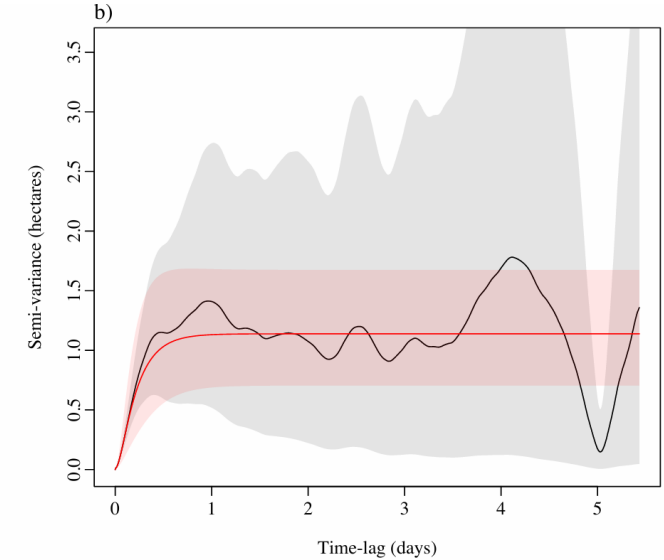
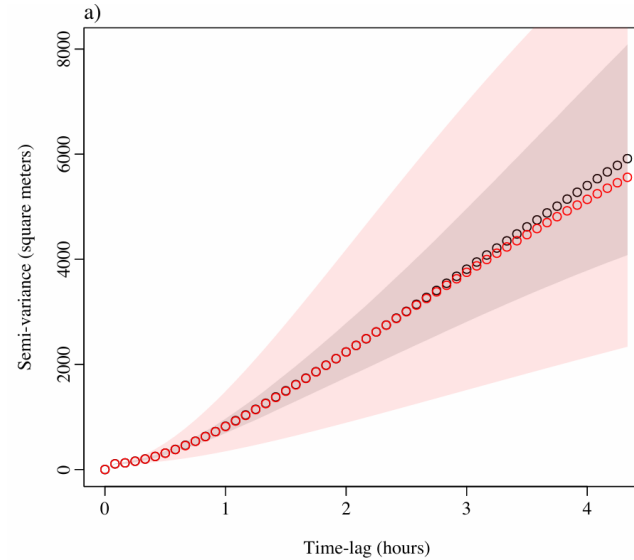


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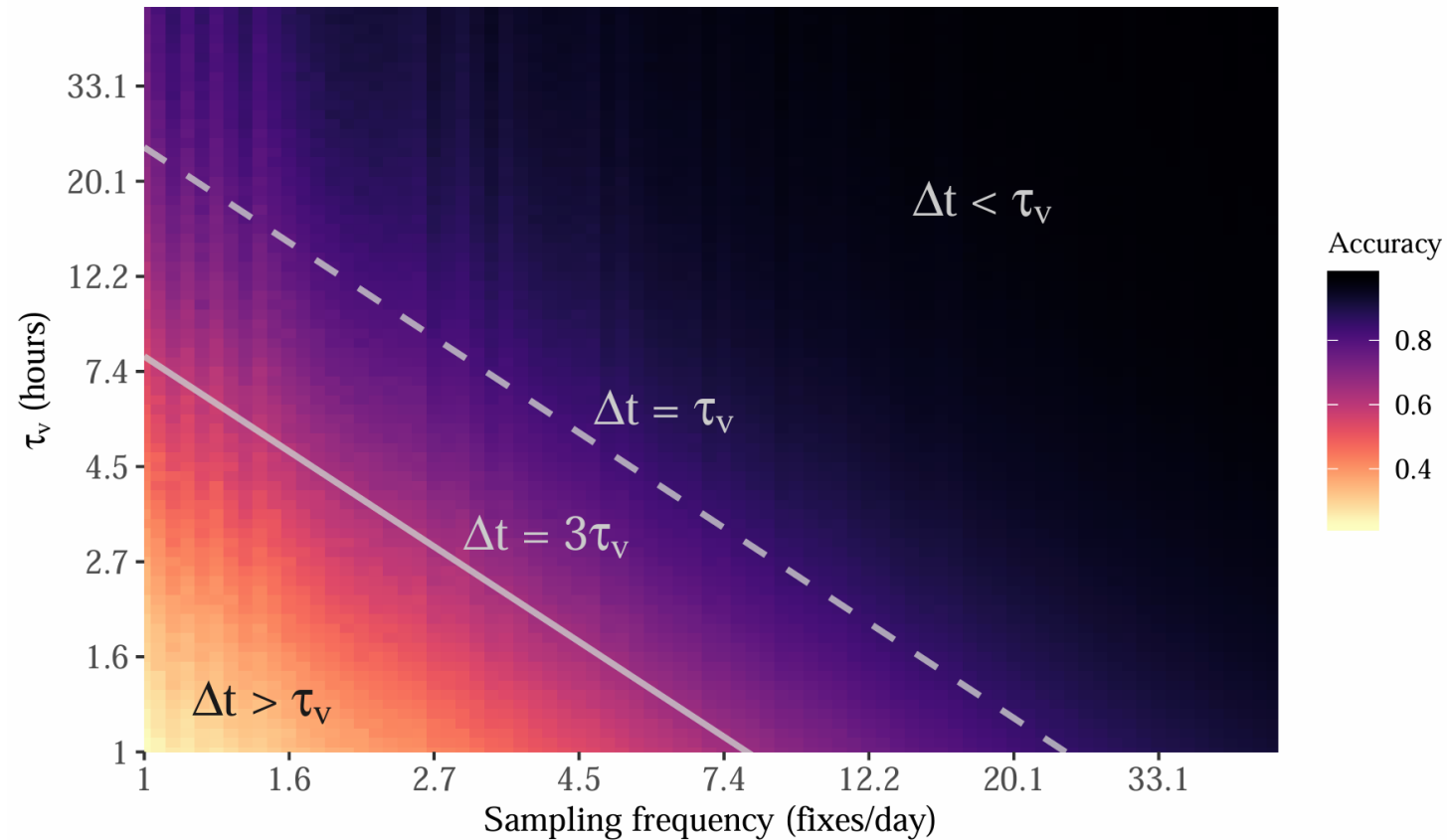
Limitations

- Conditioned on a fitted model



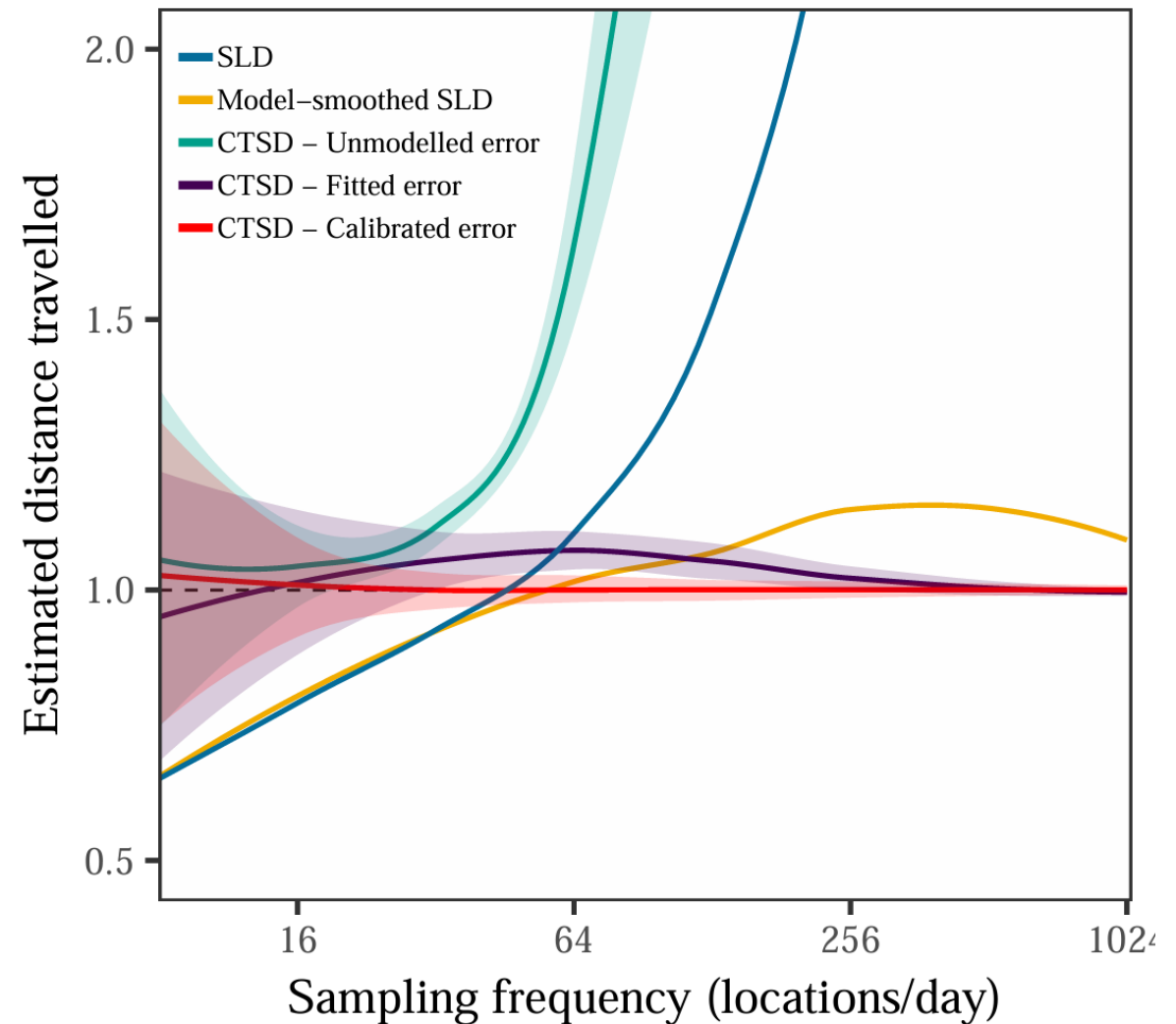
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Limitations

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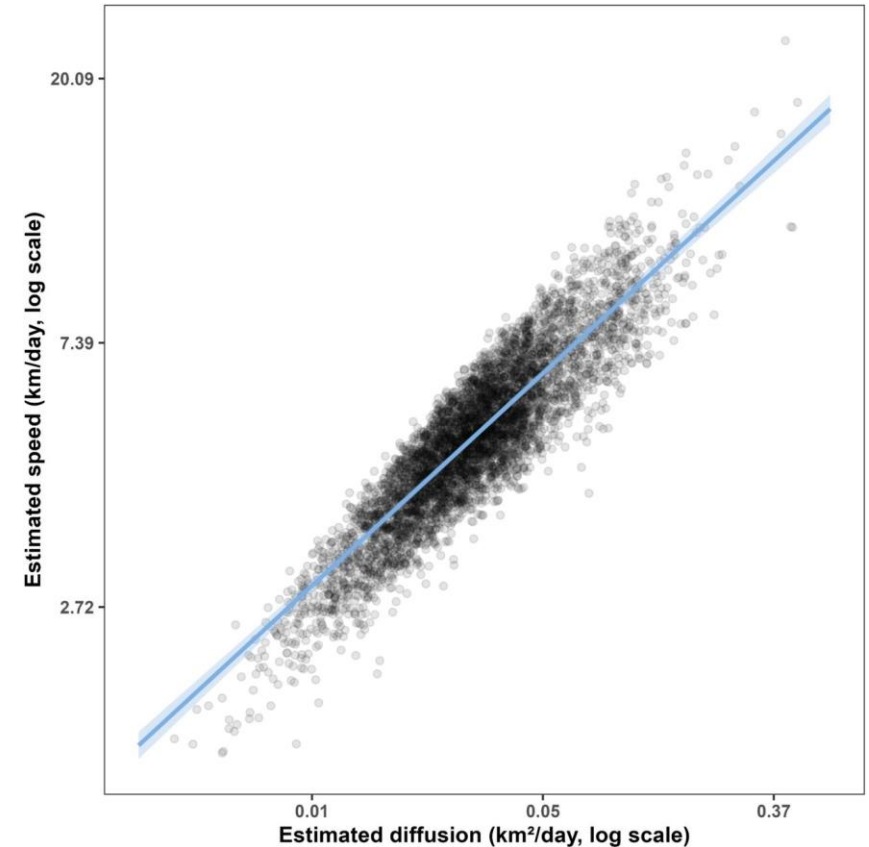


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Diffusion rate

- A more robust measure of animal **movement activity**
 - Able to be estimated accurately across a wide variety of datasets, where we may otherwise be limited for speed estimation
- Correlated with mean speed



Diffusion rate

- Reflects short-term space use to study larger-scale processes
 - Relates movement speed and tortuosity
- Higher diffusion rate → more active
 - more exploratory or dispersing
- Lower diffusion rate → less active
 - Inactive or localized activity
 - e.g., den use, foraging patch, breeding site



ctmm_speed.R

Thank you

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- Funded by and supported by:

